



Explainable train delay propagation: A graph attention network approach

Ping Huang^{a,b,c}, Jingwei Guo^d, Shu Liu^{e,*}, Francesco Corman^c

^a National United Engineering Laboratory of Integrated and Intelligent Transportation, Southwest Jiaotong University, Chengdu 610031, China

^b National Engineering Laboratory of Integrated Transportation Big Data Application Technology, Southwest Jiaotong University, Chengdu 610031, China

^c Institute for Transport Planning and Systems, ETH Zurich, Zurich 8093, Switzerland

^d Faculty of Business, City University of Macau, Macau S.A.R., China

^e Bosch Center for Artificial Intelligence, Shanghai 200335, China

ARTICLE INFO

Keywords:

Train operation data
Delay propagation
Contexts of train operation
Strengths of influences
Graph attention networks

ABSTRACT

Explaining train delay propagation using influence factors (to find the determinants) is essential for transport planning and train operation management. Due to high interpretability to train operations, graph/network models, e.g., Bayesian networks and alternative graphs, are extensively used in the train delay propagation/prediction problem. In these graph/network models, nodes represent train arrival/departure/passage events, whereas arcs describe train headway/running/dwelling processes. However, previously proposed graph/network models do not have edge weights, making them incapable of apperceiving the diverse influences of factors on train delay propagation/prediction. The train dwelling, running, and headway times vary over time, space, and train services. This potentially makes these factors have diverse strengths on train operations. We innovatively use the Graph Attention Network (GAT) to model the train delay propagation. An attention mechanism is used in the GAT model, allowing the GAT model to have arcs with diverse weights (learned from data). This enables the GAT model to discern the nodes' diverse influences; thus, with the learned importance coefficients, the model can be distinctly explained by the influencing factors. Further, the model's accuracy is expected to be improved, because the GAT model (with the attention mechanism) can pay more attention (represented by the learned weights) to the significant factors/nodes. The proposed GAT model was calibrated on operation data from the Dutch railway network. The results show that: (1) the influence factors contribute diversely to the delay propagation, and the train headway is the determinant of train delay propagation; (2) the accuracy of the proposed GAT model is significantly improved (because of the attention mechanism), compared against other state-of-the-art graph/network models. In a word, the proposed GAT method improves the interpretability of delay propagation and the accuracy of delay prediction.

1. Introduction

Railway operations are easily affected by unexpected disruptive events, e.g., human faults, equipment failure, and severe weather,

* Corresponding author at: Bosch Center for Artificial Intelligence, Shanghai 200335, China.

E-mail addresses: ping.huang@swjtu.edu.cn (P. Huang), Shu.Liu2@cn.bosch.com (S. Liu).

leading to delays in train operations and passenger traveling (Binder et al., 2017). The buffer times between trains and supplement/recovery times in sections/stations are the only resources for delay recoveries (Wang and Goverde, 2017). However, these resources are usually scarcely scheduled to guarantee transport capacity and efficiency (Feng et al., 2023). Therefore, any slight train delay may influence the trains that share the same resources (e.g., turnouts and signals), making it easy to spread to the network (Corman and Meng, 2014). In practice, train traffic controllers are required to estimate the delay propagation on the network to reduce knock-on delays or to ensure more comprehensive decisions (Huang et al., 2020b). A train delay propagation model can assist controllers in identifying more accurate delay propagation patterns and predicting future states, making it of great significance for intelligent train traffic control and delay management.

However, due to the complexity of railway systems, train delays exhibit the following characteristics (Spanninger et al., 2022): (1) **Uncertainty**: due to the stochastic characteristic of the interruptions and the variations in the utilization of the recovery times and buffer times, train delays are highly uncertain, meaning that train delays (of a single train or due to cascading propagation) in downstream can increase, decrease, or remain the same. (2) **Dynamics**: train delays are time- and space-related, inevitably showing auto-correlations, cross-correlations, and trends. (3) **Diverse train interactions**: train operations are affected by complex factors, e.g., the train running process, dwelling process, and headways. These influence factors potentially have diverse impacts/strengths on train delay propagation. Modeling the train delay propagation considering such characteristics of train delays is thus a quite challenging task. Therefore, advanced train delay propagation tools are essential in supporting the traffic controllers' decision-making by accurately estimating delay propagation and delay impacts.

The first two characteristics in train delay propagation have been well dealt with by the time-series models, e.g., the autoregression and Markov model (MM) (Wang and Work, 2015) or state-of-the-art machine learning models, e.g., the recurrent neural networks (Huang et al., 2020b), and Bayesian networks (BN) (Corman and Kecman, 2018). Although existing studies have addressed the complexity of influencing factors, see (Huang et al., 2020c; Li et al., 2022), they combined different neural networks to process the data with diverse characteristics; for example, the recurrent neural networks (RNN) were used to address time-series data, and the fully-connected neural networks (FCNN) were used to address static data. The strength-of-influence of train arrival/departure/passage events and operation running/dwelling processes are hardly discerned in the existing delay propagation or delay prediction models. In this study, **train delay prediction models** refer to the models that only care about future delays (i.e., what future train delays are). In contrast, **train delay propagation models** refer to the models caring about the propagation patterns on the railway line/network (i.e., how the delays propagate to the future). The train operation is composed of train events and running/dwelling processes, and the train delay propagation is thus driven by train events and train running/dwelling processes. A delay propagation model that can discern the strength-of-influence of these factors, is practically significant for improving the accurate detection of train delay propagation patterns, and reducing the impacts of train delays.

To this end, we propose a state-of-the-art graph-based model, Graph Attention Networks (GAT), for train delay propagation. The GAT model is built on graphs (Veličković et al., 2017), which inherits the high-interpretability advantages (i.e., using nodes to represent train arrival/departure events, rather than treating train operations as a black box) of the graph and network models in train delay propagation problems. The GAT model uses the attention mechanism (Vaswani et al., 2017) to discern the strength-of-influence of train events or processes to the future delays, which allows the delay propagation model to depend more on the events/train processes that are discerned as more critical. With the learned attention coefficients, the importance of previous states to delay propagation can be clarified, thus making train delay propagation explainable. Therefore, the novelty of this study lies in explaining train delay propagation with train events and influence factors using an attention-based graph model. The main contributions of the present study are three-fold:

- (1) Proposing a graph-based machine learning model for train delay propagation; train delays, timetable-related, and environment-related factors are considered in the proposed graph model, compared to existing graph-based delay propagation/prediction models primarily considering train delays and timetable-related factors. (e.g., the Markov model, BN model, alternative graph, and timed event graph).
- (2) Using an attention mechanism to discern the strengths of influences between nodes, allowing train delay propagation to be interpreted from both node (i.e., train arrival/departure events) and influence-factor perspectives.
- (3) The model's predictive performance and interpretability to delay propagation are improved simultaneously, compared to the existing delay propagation/prediction models that care about predictive performance or interpretability (e.g., the BN or neural network models).

The remainder of the study is structured as follows. In Section 2, the related literature is reviewed. In Section 3, the problem and method are described, and the formation and structure of the GAT model are introduced. Section 4 describes the experiments based on the Dutch railway operation data. In Section 5, the experimental results are shown and analyzed. Finally, we conclude this study and discuss the future directions in Section 6.

2. Literature review

Generally, train delays have been studied from the point of delay propagation simulation, delay propagation pattern recognition, and train delay prediction. Graph and network models have been extensively used among the proposed methods due to their high explainability/interpretability to train operations (Spanninger et al., 2022; Wen et al., 2019). The following part reviews the graph/network models used in train delay propagation and prediction problems.

(1) Delay propagation simulation

Simulation methods for delay propagation aim to mimic the delay propagation process in real-world railway systems. These methods usually need to manually set train operation disturbances or disruptions to generate primary delays (Wen et al., 2019). The most important advantage of the simulation methods is that they can approximate specific parameters that are otherwise difficult to obtain in the real world. Also, the simulation method can be easily used for sensitivity analysis of train delay propagation by the control variable method in the simulation process. However, from an offline perspective, one of their disadvantages is that the disturbances/disruptions or delays setting in these systems depend mainly on hypothetical and theoretical models (Huang et al., 2021). Some train operation simulation systems, such as OpenTrack (Nash and Huerlimann, 2004), can also be used to simulate train delay propagation. In addition, Keiji et al. (2015) formulated a train operation simulation model, which considered the interaction between the trains to reproduce the behavior of train operation. This method simulated the train operation under delay propagation (knock-on delays). Jiang et al. (2007) used a simulation method to study the characteristics of train delay propagation in the urban rail transit system, namely how service punctuality and reliability are affected by the buffer times, and reserve rolling stock quantity. Wen et al. (2016) used a train operation control system to simulate the influences of interruptions with different lengths and the delay propagation process on a high-speed railway line by investigating the number of delayed trains and the total train delays.

(2) Delay propagation modeling

Train delays easily propagate to the future stations in the networks, due to train interactions and the scarce recovery time in sections and stations. For example, a train delay easily evolves downstream (the future stations the train will visit) due to the limited recovery times in sections and stations (Huang et al., 2022; Yuan and Hansen, 2007), called **delay evolution on a single train journey (in space)**. Also, due to the train interactions, an initially delayed train easily affects its following trains. This process could lead to knock-on delays of the following trains (Carey and Kwieciński, 1994), called **delay propagations between trains (in time)**. In this regard, researchers can study the train delay propagation from the points of a single train and between trains under different scenarios (Li et al., 2021a). Recent review studies on data-driven methods show that graph- and network-based models are among the earliest and most common approaches for train delay propagation and prediction (Spanninger et al., 2022; Wen et al., 2019). All the proposed graph and network models are without aftereffects. The train arrival/departure/passing events are addressed as probabilistic to retain their randomness. The MM can interpret the delay propagation of a single train (Artan and Şahin, 2021; Barta et al., 2012; Li et al., 2021a; Şahin, 2017). The state transition probability matrix of the MM can update spatiotemporally to adapt to the variations of train running and train dwelling times (Kecman et al., 2015). The biggest disadvantage of the MM is that it can only describe a single train's operation. Other than common one-order MM, N-order MM (i.e., considering multiple past states) can be established. This means that a

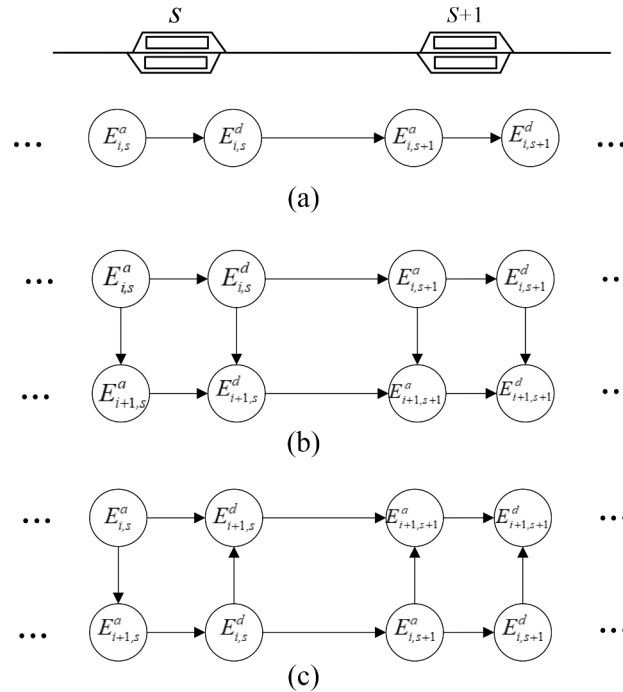


Fig. 1. The paradigms of existing train delay propagation models: (a) for a single train, (b) for multiple trains (i.e., considering train interactions) without an overtaking, and (c) for multiple trains with an overtaking, where E represents train arrival/departure events; subscripts “ i ” and “ s ” represent train and station indices, respectively; superscripts “ a ” and “ d ” represent train arrival and departure events, respectively;

state can depend on multiple past states (not just the nearest one), also called N-order (N is the number of past states considered) MM (Gaurav and Srivastava, 2018). In addition, the BN, an extension of the MM, can also be used to model the train delay of a single train. Examples include taking the past two events (e.g., arrivals or departures) as the parent nodes (Lessan et al., 2019), or taking the past events at three stations (six events) as the parent nodes (Huang et al., 2022), to establish BN models. To some extent, this paradigm/connection of the MM and BN models enables the BN and MM to capture the delay propagation patterns between successive states. The BN is built on causal inference, which means that conditioning on specific variables is one of its strengths (Corman and Kecman, 2018). Researchers focused on the train delay propagation in both time and space simultaneously instead of only focusing on train delay propagation between trains (time axis). For example, the train delay propagation in time and space was modeled by the BN model, where the authors considered two trains, and each node in the BN model depends on the previous state of the train itself, and the state of its preceding train (Corman and Kecman, 2018). The common delay propagation models in existing studies for a single train and multiple trains are shown in Fig. 1. In addition, the BN model was also used to study the influences of interruptions (primary delays, delayed trains, and total delays), where the authors built the BN model based on the inference of the three factors at different stations. Also, the timed event graph with dynamic edge weights can be used to study the delay propagation in time and space (Kecman and Goverde, 2014), where the edge weights were computed separately. The train arrival or departure delays were treated as independent variables to calculate the edge weights, whereas their nearest subsequent running times or dwelling times were addressed as dependent variables. Regression coefficients were obtained to be the edge weights of the graph model. Finally, an improved ensemble method was used to study the changes in delays from one station to the downstream stations (Zhou et al., 2020). The problem of train delay propagation/prediction can also be studied from the perspective of railway networks. In this regard, some commonly used methods include graph/network models, e.g., timed event graphs (Goverde, 2010; Hansen et al., 2010), activity graphs (Büker and Seybold, 2012), Petri nets (Milinković et al., 2013), alternative graphs (Meloni et al., 2021), and BN models (Ulak et al., 2020). In these models, a graph model that considers edge weights was proposed by (Goverde, 2010); the authors used the minimum running/dwelling process times to represent the edge weights. In addition, due to the variation in train operations, a stochastic delay propagation model has been used in German railway networks (Berger et al., 2011). Previous study shows that the distribution of secondary delays can be easily derived from the distribution of primary delays (Meester and Muns, 2007). Huang et al. proposed a decision graph model to optimize the departure order of delayed trains in merging stations, by considering delays propagation in the station (Huang et al., 2023). All the graph/network-based delay propagation models are built on discrete states (they address train delay states as nodes), retaining the high interpretability (i.e., using nodes to represent train events, rather than treating train operations as a black box) of delay propagation models. However, their main weakness is that they are incapable of addressing the diverse influences of complex factors (i.e., they cannot learn the importance coefficients from the data). For example, a new model structure (Fig. 1 (c)) is needed when a train overtaking happens (compared to Fig. 1 (b)); Further, the existing delay propagation models based on graphs and networks only take the train delays as nodes, without considering other important factors (e.g., timetable-related and environment-related factors).

(1) Train delay-related predictions

Generally, the propagation models can be used to predict train delays (Spanninger et al., 2022; Wen et al., 2019). The delay propagation methods focus on the delay propagation process from stations/sections to stations/sections. However, delay prediction

Table 1
A summary of graph/network models for train delay propagation/prediction.

Literature	Method	Considered factors	Graph type	Contains edge weight?	Research objective
(Berger et al., 2011)	MM	TD	Directed	No	Delay propagation
(Artan and Şahin, 2021)	MM	TD	Directed	No	Delay propagation
(Şahin, 2017)	MM	TD	Directed	No	Delay propagation
(Barta et al., 2012)	MM	TD	Directed	No	Delay propagation
(Kecman et al., 2015)	MM	TD	Directed	No	Delay propagation
(Gaurav and Srivastava, 2018)	MM	TD	Directed	No	Delay propagation
(Hansen et al., 2010)	TEG	TD	Directed	No	Delay propagation
(Goverde, 2010)	TEG	TD and TTF	Directed	Yes	Delay propagation
(Meloni et al., 2021)	ALG	TD and TTF	Directed	Yes	Delay propagation
(Kecman and Goverde, 2014)	TEG	TD and TTF	Directed	Yes	Delay propagation and prediction
(Corman and Kecman, 2018)	BN	TD	Directed	No	Delay prediction
(Ulak et al., 2020)	BN	TD	Directed	No	Delay prediction
(Huang et al., 2020a)	BN	TD	Directed	No	Delay prediction
(Lessan et al., 2019)	BN	TD	Directed	No	Delay prediction
(Huang et al., 2022)	BN	TD	Directed	No	Delay prediction
(Büker and Seybold, 2012)	ACG	TD	Directed	No	Delay propagation
(Milinković et al., 2013)	PN	TD	Directed	No	Delay prediction
(Zhang et al., 2021)	GNN	Multiple factors	Undirected	Yes	Delay prediction
(Zhang et al., 2022)	GNN	Multiple factors	Undirected	Yes	Delay prediction
This study	GAT	TD, TTF, and EF	Undirected	Yes	Explainable delay propagation and prediction

Note: Markov model (MM), timed event graph (TEG), alternative graph (ALG); Petri net (PN), Bayesian networks (BN), activity graph (ACG), graph neural networks (GNN), graph attention networks (GAT). Train delays (TD), timetable-related factors (TTF), environment-related factors (EF).

methods usually care about the length of future train delays, aiming at providing the traffic controllers with an estimation of future train delays (Shi et al., 2021). The state of the art of prediction methods can use historical data, such that the learned knowledge and phenomena can be applied to a new situation. Generally, the state-of-the-art event-driven methods (e.g., alternative graph, activity graph, and timed event graph), BN models, and graph neural networks (GNN) have been widely used in train delay-related prediction problems based on train operation data. For example, the BN model has been used in various train delayed-related issues, e.g., train delay forecasting/prediction (Corman and Kecman, 2018; Huang et al., 2022; Lessan et al., 2019), the estimation of the influence of railway interruptions (Huang et al., 2020a), the prediction of train delay's impacts from a network perspective (Ulak et al., 2020), and disruption length prediction (Zilko et al., 2016). Recently, the GNN model, due to its interpretability compared to traditional machine learning models (e.g., recurrent neural networks and convolutional neural networks), has been applied to the train delay prediction problem (Heglund et al., 2020; Zhang et al., 2021), and the delays of train operations in stations (Zhang et al., 2022).

A summary of the graph/network-based models for train delay-related problems is shown in Table 1. Dominant studies on graph/network models indicate the model's interpretability to train operations. Although many types of graph/network-based models have been proposed for railway delay propagation/prediction problems, they have one or more of the following issues:

- (1) **Only built on train delays and timetable-related factors.** Dominant existing models were only built on train delays and timetable-related factors, without considering other important factors (Corman and Kecman, 2018; Hansen et al., 2010; Şahin, 2017). Environment-related factors, like rainfall and wind speed, are essential to train delay propagation, as they can somewhat influence delay propagation.
- (2) **Incapable of addressing diverse impacts of factors.** Only two studies based on traditional methods (e.g., the BN and timed event graphs) considered edge weights, representing the impacts of factors, but the edge weights were roughly determined (Goverde, 2010; Kecman and Goverde, 2014). For example, actual running/dwelling times were exploited as edge weights in (Goverde, 2010); in addition, regression coefficients were used as edge weights in (Kecman and Goverde, 2014); the regression coefficients were obtained by taking rain delays as the independent variables, and actual train running and dwelling times as the dependent variable.
- (3) **Inexplainable for delay propagation.** Most studies only focus on delay prediction. The proposed graph or network models are inexplainable to delay propagation, e.g., how much a factor contributes to delay propagation (Zhang et al., 2021; Zhang et al., 2022).
- (4) **The structure is inflexible to train overtaking.** The previously proposed methods were mainly built on directed graphs (e.g., the timed event graphs and BN), meaning that the model structure needs to be adjusted when a train overtaking happens (Corman and Kecman, 2018; Huang et al., 2020a; Kecman and Goverde, 2014).

In the present study, we propose a GAT model for train delay prediction and propagation problem, which considers train delays, train running, dwelling, and headways and learns the edge weights based on input data using an attention mechanism (Vaswani et al., 2017). The GAT model allows both unidirectional and bidirectional graphs. Therefore, we used a bidirectional graph to construct the delay propagation model. The structure of the proposed GAT model does not need to be changed under train overtaking conditions, thus improving the practicability of the proposed delay propagation model. The bidirectional graph is understandable, given that the traffic controllers may not only consider the past information (e.g., delays in the upstream stations), but estimate the future information (e.g., delays at downstream stations) to make a decision. Also, the GAT can learn an importance coefficient (weight) for each edge, allowing us to explain delay propagation from a micro perspective. In addition, using the GAT model's graph structure, the proposed model can retain the advantages (high interpretability) of traditional train delay propagation models. The proposed model is expected to improve the interpretability of delay propagation and the accuracy of delay prediction.

3. Problem statement and method

3.1. Problem statement

Railway networks/lines are composed of stations and sections. In this study, sections refer to the tracks between two successive stations. The sections have different lengths, leading to diverse train running times. Further, trains have different stopping patterns; for example, local trains may stop at every station, but long-distance trains stop at major stations. The diverse stopping patterns make the train running times in a section different (acceleration and deceleration). In addition, train dwelling times differ from services and stations, mainly depending on the passenger amount. Most importantly, train running times in sections (usually take a few minutes to tens of minutes) also distinctively differ from dwelling times in stations (usually take less than five minutes); train running processes in sections usually take longer time than train dwelling processes in stations. For example, high-speed trains take seven to nineteen minutes in sections, whereas they take one to six minutes in stations (Wen et al., 2017). Dutch local trains take minutes between two consecutive stations, whereas they only take no more than one minute at stations (Kecman and Goverde, 2015). Additionally, the diverse travel demand in different periods of the day makes the headways of train services different. For instance, trains may operate with quite short headways (even with the minimum headway) in peak hours. Trains operations with short headways mean less robust timetables, because of fewer buffers and recovery times, making it difficult for delays to be recovered (Fischetti et al., 2009). In contrast, train delays can be easily recovered under long-headway conditions.

The differences in train running, dwelling, and headway times stated above indicate that train events have diverse strengths of interactions/dependencies in delay propagations. The various strengths of influences in train operations are shown in Fig. 2. In this

figure, the thickness of the arrows represents the strength of influences. For example, a train with longer delays may have stronger influences on its succeeding trains, meaning that the connections between these trains should be stronger than normal trains (edges are thicker). Due to their interpretability to train operations, graph/network models are widely used for train delay propagation/prediction; see the two review papers (Spanninger et al., 2022; Wen et al., 2019). However, most traditional graph and network models, e. g., Bayesian networks and Petri net, cannot address the various dependencies between train events in prediction tasks. Although two studies took edge weights into account (Goverde, 2010; Kecman and Goverde, 2014), the determination of the edge weights highly depends on domain assumption (e.g., the running time is only affected by departure delays, and the dwelling time is only influenced by arrival delays). The diversities of train headways, running times, and dwelling times indicate that more advanced models that can address complex factors and the diverse influence of factors are needed. Further, another disadvantage of the existing delay propagation models (shown in Fig. 1) is that the structure of the model needs to be updated when a train overtaking happens. This is mainly because the traditional delay propagation models (e.g., MM or BN) are unidirectional graphs.

To address these problems, we use a state-of-the-art graph-based model with an attention mechanism called graph attention networks (GAT), to study the delay propagation problem. The GAT model, built on bidirectional graphs, can learn the strengths of the edges between nodes from the historical train operation data by the attention mechanism. These qualities enable the GAT model to address the train overtaking problem without any structure change in real-time, and to pay more attention to those nodes with which the connections are stronger. The bidirectional graph is appropriate since the traffic controllers may consider their previous actions and future possible actions to decide. In addition, the attention mechanism learns the edge weights from the data, avoiding defining the edge weight by users. In the train delay prediction/propagation problem, the ability to address train overtaking can guarantee higher practicability of the model; the strength-of-influence learning ability enables the future train events (or future train delays) to have different dependencies on the previous train events, thus providing a more accurate modeling of the delay propagation and more detailed information for train traffic controllers about what to pay their attention to.

3.2. Influence factors

As mentioned in Section 3.1, recovery/supplement/buffer times are different from stations/sections/services, making the strengths of influences on train delay propagation diverse. To discern the diverse strengths of influences between nodes, we considered train delays, planned timetable-related, and actual timetable-related factors in the delay propagation model. In addition, we also considered the influences of the environment. Two critical indexes were selected as the influence factors of the delay propagation model: rainfall and wind speed. The definitions of the selected influence factors are as follows:

(a) Features in terms of train delays (FTD):

- Arrival delays (T^a):

$$T_{n,s}^a = at_{i,s}^a - pt_{i,s}^a, at_{i,s}^a \in AE^a, pt_{i,s}^a \in PE^a \quad (1)$$

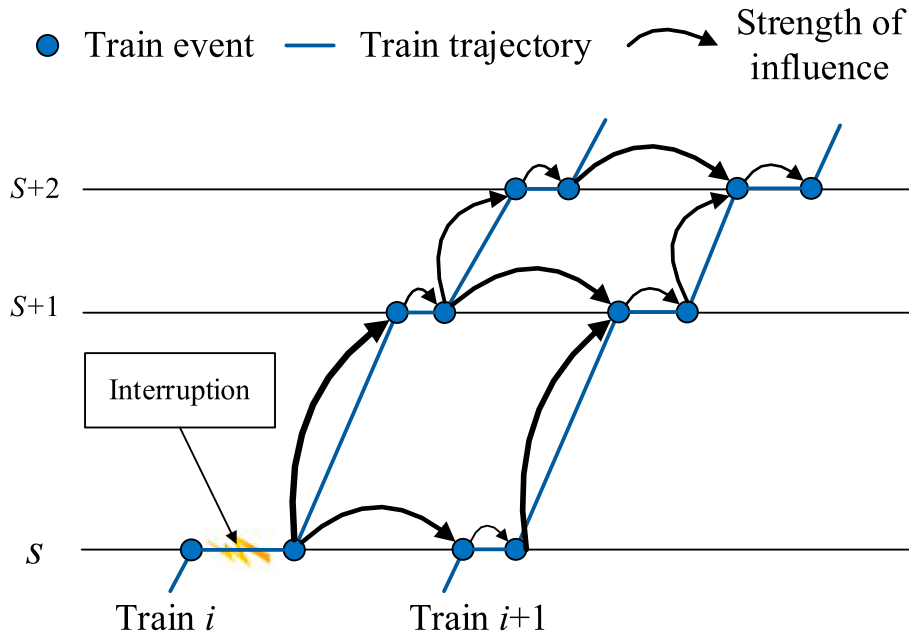


Fig. 2. The influences between train events in train operations.

- Departure delays (T^d):

$$T_{i,s}^d = at_{i,s}^d - pt_{i,s}^d, at_{i,s}^d \in AE^d, pt_{i,s}^d \in PE^d \quad (2)$$

Where subscripts “ i ” and “ s ” represent train and station indices, respectively; superscripts “ a ” and “ d ” represent train arrival and departure events, respectively; $at_{i,s}^a$ and $pt_{i,s}^a$ represent the actual and planned arrival times of train i at station s , respectively; $at_{i,s}^d$ and $pt_{i,s}^d$ represent the actual and planned departure times of train i at station s , respectively; AE^a , PE^a , AE^d , PE^d are the sets of actual arrival events, planned arrival events, actual departure events, and planned departure events in the timetable, respectively.

(2) Features in terms of the planned timetable (FPT):

- Planned dwelling time (PDT):

$$PDT_{i,s} = pt_{i,s}^d - pt_{i,s}^a, pt_{i,s}^d \in PE^d, pt_{i,s}^a \in PE^a \quad (3)$$

- Planned running time (PRT):

$$PRT_{i,s \rightarrow s+1} = pt_{i,s+1}^a - pt_{i,s}^d, pt_{i,s}^a \in PE^a, pt_{i,s}^d \in PE^d \quad (4)$$

- Planned arrival headway (PH^a)

$$PH_{i-1 \rightarrow i,s}^a = pt_{i,s}^a - pt_{i-1,s}^a, pt_{i,s}^a \in PE^a, pt_{i-1,s}^a \in PE^a \quad (5)$$

- Planned departure headway (PH^d)

$$PH_{i-1 \rightarrow i,s}^d = pt_{i,s}^d - pt_{i-1,s}^d, pt_{i,s}^d \in PE^d, pt_{i-1,s}^d \in PE^d \quad (6)$$

(a) Features in terms of the actual timetable (FAT):

- Actual dwell time (ADT):

$$ADT_{i,s} = at_{i,s}^d - at_{i,s}^a, at_{i,s}^d \in AE^d, at_{i,s}^a \in AE^a \quad (7)$$

- Actual running time (ART):

$$ART_{i,s \rightarrow s+1} = at_{i,s+1}^a - at_{i,s}^d, at_{i,s}^a \in AE^a, at_{i,s}^d \in AE^d \quad (8)$$

- Actual arrival headway (AH^a):

$$AH_{i-1 \rightarrow i,s}^a = at_{i,s}^a - at_{i-1,s}^a, at_{i,s}^a \in AE^a, at_{i-1,s}^a \in AE^a \quad (9)$$

- Actual departure headway (AH^d):

$$AH_{i-1 \rightarrow i,s}^d = at_{i,s}^d - at_{i-1,s}^d, at_{i,s}^d \in AE^d, at_{i-1,s}^d \in AE^d \quad (10)$$

(a) Features in terms of environment (FE):

- Wind speed (WS): the maximum wind speed in a time unit was used (unit: m/s). Here, the time unit is 1 h, meaning that the wind speed was calculated hourly.
- Rainfall (RF): the total rainfall in a time unit (unit: mm). Here, the time unit is 1 h, meaning that the total rainfall was in an hour.

These factors will be the input of the proposed GAT model, to capture their influences on train delay propagation.

3.3. The self-attention

As shown in Fig. 2, a state-of-the-art model is needed to capture the diverse interactions between train events. The attention mechanism is a technology enabling a model to focus on important information. For example, the graph model can learn an edge weight for each pair of neighboring nodes based on the node features (Vaswani et al., 2017). Given a train delay propagation model built on train events (like Fig. 1), the diverse influences of train events on delay propagation patterns can thus be captured by self-attention.

The self-attention mechanism is composed of three main parts, i.e., the keys (**K**), the queries (**Q**), and the values (**V**). Node features can express these three components in the train delay propagation model. Let us assume an input (of the input layer or hidden layers), $H_{N,M} = \{h_{1,M}, h_{2,M}, \dots, h_{N,M}\}$, where $h_{n,M}$ represent a vector, M is the length of the vector, i.e., the number of features of a node, and N is the data volume, n and m are the sample and feature indices, $n \in \{1, 2, \dots, N\}$, $m \in \{1, 2, \dots, M\}$. The attention mechanism can be implemented with the following steps (Vaswani et al., 2017):

(1) The **Q**, **K**, and **V** matrices are first generated by multiplying the input H and three learnable weight parameters W^q , W^k , and W^v , respectively, to obtain the linear maps of **Q**, **K**, and **V**, shown as (Suppose matrices W^q , W^k , and W^v have a dimension of $M \times P$):

$$Q_{N,P} = H_{N,M} W^q \quad (11)$$

$$K_{N,P} = H_{N,M} W^k \quad (12)$$

$$V_{N,P} = H_{N,M} W^v \quad (13)$$

Where $Q_{N,P}$, $K_{N,P}$, $V_{N,P}$, are linear maps of **Q**, **K**, and **V**. For neural network models, the weights between neurons, i.e., W^q , W^k , and W^v , are usually randomly initialized using, e.g., Gaussian distribution. In practice, a bias vector may be added to the product of matrix multiplications.

(2) Compute the attention coefficient ($a_{x,y}$) between nodes x and y . The **query** vector (**Q**) is matched against all **key** vectors (**K**) to compute the score value, by performing a concatenation operation, shown in (4):

$$e_{x,y} = A(Q_{N,P}, K_{N,P}) \quad (14)$$

Where “A()” represents an attention operation, which could be an end-to-end concatenation or dot product operation; an end-to-end concatenation operation of vectors is to place vectors side-by-side to form one vector, while the length of the new vector is the summation of the previous vectors’ lengths.

(3) The score values are then passed through an exponential function, expressed by “exp()”, and normalized to obtain the attention coefficient:

$$a_{x,y} = \frac{\exp(e_{x,y})}{\sum_{n=1}^N \exp(e_{x,n})} \quad (15)$$

where $a_{x,y}$ is the attention coefficient between node x and node y .

3.4. Graph attention networks (GAT)

In general, a graph (G) can be represented by $G = (E, R)$, where E represents a set of nodes in G , and R represents a set of edges connecting two different nodes (West, 2001). A graph can be directed (edges have orientations) or undirected (edges do not have directions). The nodes of traditional delay propagation models usually describe the train arrival/departure/passage events at stations. In contrast, arcs between nodes represent the train running/dwelling/headway processes. One of the disadvantages of the traditional graphs (e.g., BN) is that they cannot recognize the diverse influences of the neighbor nodes, meaning that they treat each node equally crucial in a prediction task.

GATs are used in the present study. GATs are neural networks built on graphs with an attention mechanism to determine critical information (Velićković et al., 2017). The attention mechanism aims to multiply a coefficient for each node when transmitting information on the graph. Therefore, the advantage of using the GAT model in delay propagation problems lies in its attention mechanism, which allows the model to pay more attention to essential train events. In comparison, other standard graph models for train operations, e.g., graph convolutional networks (Zhang et al., 2021) and Bayesian networks (Huang et al., 2022) need manually set edge weights or are incapable of addressing the different strength-of-influence of upstream train operation events/processes in prediction tasks.

In a GAT layer, each node in the graph can assign different weights according to the characteristics of other nodes, i.e., the features of train operations. Suppose the input to a graph attention layer is a set of node features, $H = \{h_1, h_2, \dots, h_{\bar{E}}\}$, $h_d \in \mathbb{R}^M$, $d \in \{1, 2, \dots, \bar{E}\}$, where d is the node index, $\bar{E}$ is the number of nodes in the graph, and M is the number of features of each node. In a train delay propagation model, the input features of a node can be the train delay, planned running times, and planned dwelling times. Then, the layer produces a set of node features, $H' = \{h'_1, h'_2, \dots, h'_{\bar{E}}\}$, $h'_d \in \mathbb{R}^M$, $d \in \{1, 2, \dots, \bar{E}\}$, M is the number of features of each node in the outputs, which is potentially different from the number of features of each node in the inputs. A shared linear transformation (i.e., a linear map between two vector spaces preserving the operations of vector addition and scalar multiplication), parametrized by a weighted matrix, $W \in \mathbb{R}^{F \times F}$, is first applied to each node to obtain a higher-level representation of the input features. The output features can be obtained based on the input features by:

$$h'_x = a_{x,x} W^v h_x + \sum_{d \in B(x)} a_{x,d} W^v h_d, d \in \{1, 2, \dots, \bar{E}\} \quad (16)$$

Where d is the node index, “ $B(x)$ ” represents the neighbor nodes of node x . Therefore, the attention score ($e_{x,y}$) of nodes x and y can be

calculated by Eq. (14). To make coefficients easily comparable across different nodes, $e_{x,y}$ are normalized across all neighbor nodes by:

$$a_{x,y} = \frac{\exp(e_{x,y})}{\sum_{d \in B(x)} \exp(e_{x,d})} \quad (17)$$

where $a_{x,y}$ is the normalized attention coefficient between nodes x and y .

Therefore, calculating an attention coefficient between node x and node y is first to perform linear transformations on both nodes. Then, an attention operation, e.g., a concatenation or a dot product operation, will be performed to calculate the raw attention coefficient, representing node y 's importance to node x . Finally, the normalized attention coefficient can be obtained by performing a normalization operation across all neighbor nodes of node x . Thus, the attention coefficients between node x and all the neighbor nodes can be computed. This whole process can be seen in Algorithm 1. In a GAT graph, the attention coefficient will be calculated for each node and its neighbor nodes, thus obtaining attention coefficients for each edge.

According to Ref (Velicković et al., 2017), an illustration of calculating the attention coefficient between two nodes x and y is shown in Fig. 3(a). The GAT model enables the delay propagation model to recognize the context of neighbor nodes. In addition, multiple attention coefficients can be calculated for each edge in the GAT model, to strengthen the performance of the GAT model in extracting the critical features. The calculation of each attention coefficient is constant, shown in Fig. 3(a). After calculating multiple attention coefficients, the final attention coefficient can be obtained by applying an average or end-to-end concatenation operation. The illustration of multiple attention coefficients is shown in Fig. 3(b).

Algorithm 1. (The calculation of the attention coefficient in a GAT layer)

Input: A set of node features $H = \{h_1, h_2, \dots, h_{\bar{E}}\}$, $h_d \in \mathbb{R}^M$, $d \in \{1, 2, \dots, \bar{E}\}$, where d is the node index, $\bar{E}$ is the number of nodes, and M is the number of features in each node. The neighbor nodes of node x are represented by $B(x)$.

Output: the average attention coefficient between nodes x and y , $a_{x,y}$, $y \in B(x)$.

for $d = 1, 2, \dots, \bar{E}$ **do**:

if $d \in B(x)$ **do**:

 perform a linear transformation on the node features h_x and h_d , $d \in B(x)$, parametrized by a weighted matrix W^q and W^k , respectively, to obtain two new representations of the node features: $W^q h_x$, and $W^k h_d$;

 Apply an attention operation "A()" on $W^q h_x$, and $W^k h_d$ using Eq. (14), and obtain $e_{x,d}$;

end if

end for

Using Eq. (17) to obtain the normalized attention coefficient ($a_{x,y}$) between node x and y , $y \in B(x)$.

3.5. The graph paradigm for delay propagation

The traditional graph- and network-based models for train delay propagation were established by interpreting train operations graphically, where the nodes of the graph/networks represent the train arrival and departure events at stations. The connections/arcs between nodes describe the relationships (i.e., running and dwelling process or headway times) between train events. This study also uses the most standard way to implement the delay propagation model in the existing references (Corman and Kecman, 2018; Huang et al., 2020a; Kecman and Goverde, 2014; Lessan et al., 2019). To determine the structure of the model for delay propagation, we

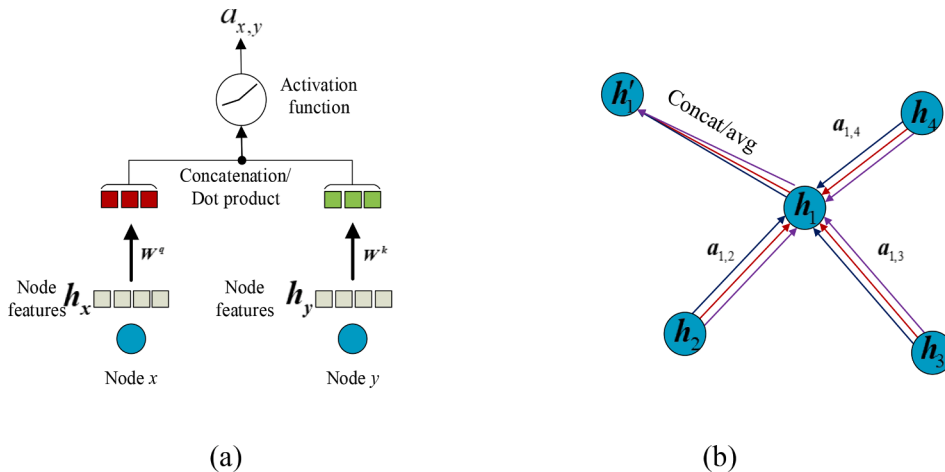


Fig. 3. The calculation of attention coefficient (a) and multi-head attention (b).

borrowed the paradigm proposed by (Corman and Kecman, 2018). In this study, the delay propagation model was built on the operation events of consecutive trains to consider train interactions. In the delay propagation model, each departure event of a train depends on its arrival event at the same station; each arrival event of a train depends on its departure event at the previous station. To consider the interactions/dependency between trains, the preceding train's arrival and departure events influence the succeeding train's corresponding train events. However, the existing models are all built on directed graphs, requiring the mode structure to be changed accordingly when a train overtaking happens.

The GAT model allows both unidirectional and bidirectional graphs. We used a bidirectional graph to construct the delay propagation model, which guarantees a more straightforward procedure in real-time delay prediction. This is because the structure of the proposed GAT model does not need to be changed under the train overtaking condition, compared to the traditional unidirectional graph models whose edge direction needs to be reversed (i.e., a new model is needed), due to train overtaking. Therefore, the number of unidirectional graph models needed is proportional to the number of overtaking among the delayed train group. The bidirectional graph, on the one hand, can perform a quicker response by always using just one structure under train-overtaking conditions and, on the other hand, can improve the generalizability of the train delay propagation model. However, this advantage of the bidirectional graph mainly applies to the railway line with dense traffic. For those railway lines with long train headways (e.g., longer than 20 min), there may be no train overtaking or delay propagation under interruptions. Unidirectional graphs can be more appropriate, as there is no need to update structure in real-time traffic management.

The bidirectional graph is understandable, given that traffic controllers may not only consider the past parameters (e.g., delays in the upstream stations), but estimate the future states (e.g., delays at downstream stations) to make a decision. For instance, if train $i + 1$ overtakes train i at station s , the connections of the two trains at arrival events should be $P(E_{i+1}^a | E_i^a)$, and those of the two trains at departure events should be $P(E_i^d | E_{i+1}^d)$, expressed by conditional probabilities. Changing the model structure inevitably increases the model's complexity and lowers its usability in real-time prediction. The GAT used in the study considers the influences of two nodes on each other; the connections between two consecutive trains can be constant (shown in Fig. 4) from the origin to the destination station. When an overtaking happens, the learned attention can describe the dependency between nodes. In GAT, the attention mechanism will learn two attention scores for a bidirectional arc, each direction with an attention score.

The structure of the proposed delay propagation model is shown in Fig. 4. It should be noted that more trains and nodes can be added to the graph to consider train interactions. Experiments will be designed to select the number of trains considered in the following part. The delay propagation model in Fig. 4 was built on three assumptions: (1) train interactions exist among trains, and consecutive trains have dominant interactions; (2) The influences of delay causes on train delay propagations are negligible; in other words, future train delays are dominantly influenced by previous and current train delays; (3) in real-time train traffic control, traffic controllers not only consider current single state, but the past or the future train operation states when making decisions. These three assumptions are intuitive in railway train operations, and can be easily found in previous studies (Corman and Kecman, 2018; Ge et al., 2023; Goverde, 2010; Huang et al., 2021; Huang et al., 2020b; Kecman and Goverde, 2014). In (Huang et al., 2021; Huang et al., 2020b), the authors proved that train delays depend on previous trains. They showed that the performance of a propagation/prediction model significantly improved, by considering the train interactions through experiments. In (Corman and Kecman, 2018; Goverde, 2010; Kecman and Goverde, 2014), the author also assumed that the influences of delay causes on delay propagations are negligible; they only built train delay propagation models on train delays and train operation data. In (Ge et al., 2023), the authors addressed the train traffic control actions (e.g., retiming trains) as sequences, and modeled the cascading effects between traffic control actions by Bayesian networks. The results showed that traffic control actions are dependent. These indicate the rationality of our assumptions.

Finally, traditional graph models either only consider train delays or consider train delays and timetable-related factors. We considered train delays, timetable-related factors, and environment-related factors. Fig. 4 just shows the connection of train events. Each node may contain multiple features (see Section 3.2). In addition, because the proposed GAT model is performed based on a graph regression problem (similar to graph classification) (Velicković et al., 2017), where the input is the entire graph, and the output is the

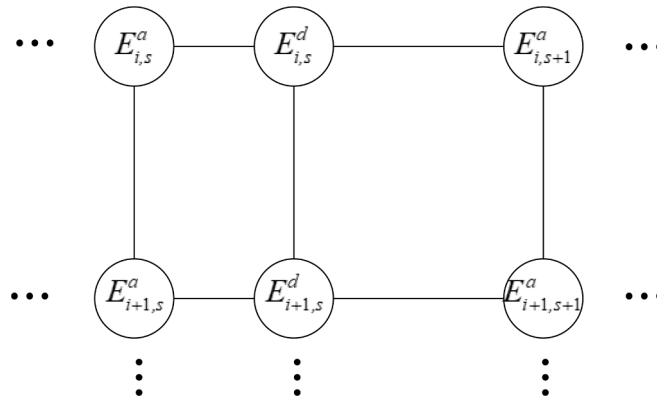


Fig. 4. A possible graph structure for delay propagation derived from existing studies, where E represents train arrival/departure event; superscripts a and d represent arrival and departure events, respectively; subscripts i and n represent train and station indices, respectively.

future train delay, the bidirectional edges will not influence the prediction horizon (i.e., time in advance) of the model. The GAT model built on Fig. 4 will learn the attention coefficients (edge weights) from the data, where the calculation of the attention coefficient can be expressed by Algorithm 1.

4. Experiments for model selection

4.1. Data description and analyses

(1) data description

The proposed GAT model was tested on operational data from a railway line from the Dutch railway network, namely the mainline line from Amsterdam to Utrecht, called the A-U line. The sketch map of this railway line is shown in Fig. 5. A maximum train speed of 140 km/h is allowed on the A-U line. Three types of trains operate on the line, including the LM train, IC train, and SPR train, in a proportion of 47: 3: 50. Train services from Amsterdam to Utrecht were used. The data contains actual/planned arrival and departure times of trains at each station, service dates, and train numbers. We focused on the section from Asb to Bkl.

The train operation data was collected at checkpoints (sensors located). The planned and actual arrival and departure times of each train, rainfall, and wind speed at each checkpoint were collected. The train operation data ranges from September 4th, to December 8th, 2017 (only on weekdays). Then, the influence factors in (Section 3.2) were calculated from the dataset. Next, some data pre-processing techniques were applied to the data, including removing trains that operated at midnight (0:00–6:00, these trains had long headways with other trains, and delay propagation was rare on these trains), filling missing cells and cells with errors (e.g., negative dwelling or running times) with the mean value. It is worth noting that train delays have been proven to be unbalanced data (long delays have lower frequencies) (Lessan et al., 2019). This data quality easily leads to a biased model with lower performance on less-frequent samples.

After data preprocessing, the collected data contains 9,584 train services. To avoid over-fitting, we randomly separate the whole data into training (accounting for 70 %) and testing datasets (accounting for 30 %), resulting in 6,708 cases/trains and 2,876 cases/trains for training and testing, respectively. In the model training process, we randomly separated the training data into two parts, 80 % for parameter learning, and 20 % for validation.

(2) delay propagation investigation based on data analysis.

Delay jumps are the main measurement of delay propagation patterns. A delay jump means a delay increase or decrease, influencing the distance a delay can propagate. For example, when the recovery time/buffer time is used to absorb the delays (i.e., the delay decreases), it leads to a delay jump. To better know the delay propagation patterns, we first investigated the distributions of delay jumps over the three main processes (i.e., dwelling, running, and headways). The calculations of the delay jump for these three processes are shown below.

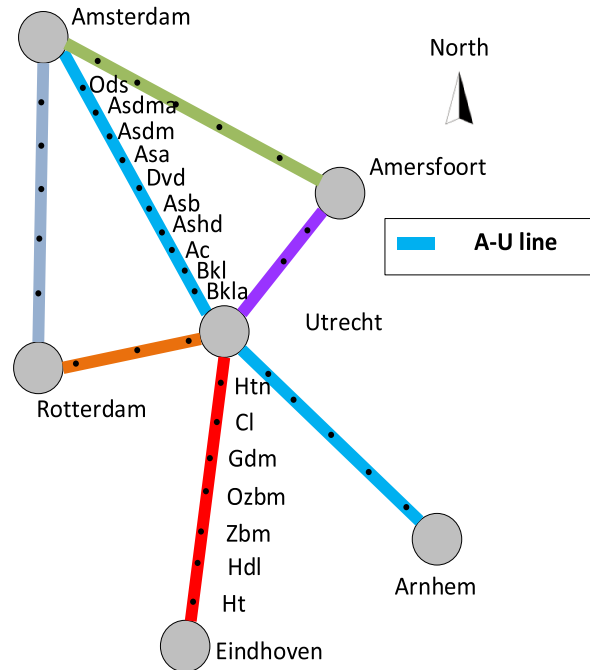


Fig. 5. The layout of the Dutch railway line.

$$U_{dew} = T_{i,s}^d - T_{i,s}^a, \quad (18)$$

$$U_{run} = T_{i,s}^a - T_{i,s-1}^d \quad (19)$$

$$U_{hea} = T_{i+1,s}^a - T_{i,s}^a, \text{ or} \\ = T_{i+1,s}^d - T_{i,s}^d \quad (20)$$

where U_{dew} , U_{run} , and U_{hea} are the delay jumps over the dwelling, running, and headways.

We show the distributions of the absolute delay jumps of the three processes in Fig. 6. In Fig. 6, the solid red line represents the mean value. It clearly shows that, compared to the delay jumps of the dwelling and running processes, the delay jumps over the headway process are the most significant since their mean value and variance are substantially larger than their counterparts. This result also demonstrates the strongest influences of the headways on delay jumps (i.e., delay propagation). In the present study, we use the GAT model to learn this domain knowledge to contribute to delay propagation pattern identification and future delay prediction.

4.2. Experiments for hyperparameter selecting

To improve the performance of the GAT model, its parameters were tuned based on the cross-validation method. We used 80 % of the training data for parameter learning, and the remaining for validation. We optimized six critical parameters of the GAT model, including the number of GAT layers, the number of GAT units in each layer, the number of attention heads, the dropout rate, the batch size, and the learning rate. Two standard metrics for regression models, i.e., mean absolute error (MAE), and root mean squared error (RMSE), as shown in (21)-(22), are used as the metrics to show the performances of the GAT model on experiments.

$$MAE = \frac{1}{N} \sum_{j=1}^N |f_j - g_j| \quad (21)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{j=1}^N (f_j - g_j)^2} \quad (22)$$

where N is the data volume, f_j is the actual value, and g_j is the estimated value.

We set appropriate candidate values for each parameter, and repeated each experiment 10 times. The number of GAT layers was selected from 1, 2, 3, and 4; the number of GAT units per layer was selected from 8, 16, 32, and 64; the number of attention heads was selected from 8, 16, 32, 64, and 128; the dropout rate was selected from 0.0, 0.1, 0.2, 0.3, and 0.4; the batch size was selected from 16, 32, 64, and 128; the learning rate was selected from 10^{-6} , 10^{-5} , 10^{-4} , 10^{-3} , and 10^{-2} . The experimental results (mean values of 10 runs) based on the cross-validation technique are shown by the line plots in Fig. 7. It can be seen from Fig. 7 that each parameter has observable influences on the model's performance. The learning rate is sensitive to the model's performance, where a few changes in the learning rate can lead to a massive fluctuation in the model's performance. In comparison, the dropout rate has the most ignorable influences on the model's performance. It can be seen from Fig. 7 that the performance of the GAT model improves with the increase of the GAT layers, GAT units per layer, and the number of attention heads. The loss value will reach a plateau when these three parameters further grow. This is because too simple a network structure easily makes the GAT model under-fitting. However, the dropout

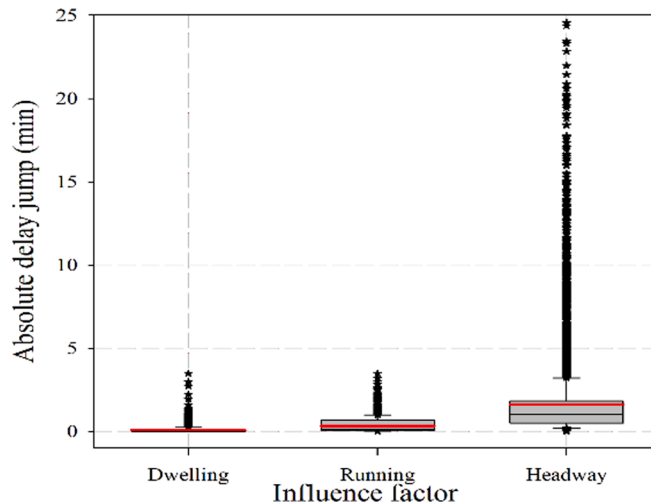


Fig. 6. The distribution of delay jumps for dwelling, running, and headways.

rate, batch size, and learning rate produce a basin shape in the loss curve, meaning that too low and too high values of these three parameters could both harm the model's performance. Therefore, an appropriate selection of these parameters is needed before using the model.

Based on the results in Fig. 7, three layers, each with 32 units, 64 attention heads, dropout rate = 0.1 (disconnecting 10% of the network nodes), batch size = 32, and learning rate = 0.0001 were selected as the model's parameters.

To show the influence of parameters on the model's efficiency, we also displayed the training time of each parameter setting, shown by the bars in Fig. 7. They indicate that the GAT model's efficiency is sensitive to the number of GAT layers, the number of units per layer, and batch size. A subtle change in these three parameters led to significant fluctuations in model training time. This is mainly because these parameters affect the structure/complexity of the model or the number of data batches. The number of attention heads, dropout rate, and learning rate had ignorable influences on the GAT model's training time, as they have subtle relationships with the model's complexity.

4.3. Experiments for feature assignments

We selected the running times, dwelling times, and headways as the input to capture the influences of the timetable-related factors on delay propagation. The proposed GAT model has both node attributes/features and edge attributes/features. This means that the selected factors can be addressed in two ways: (1) using both node features and edge features, i.e., treating train delays as node features, and treating other factors as edge features; (2) only node features, i.e., treating all features as node features. The first case (node feature + edge feature) is intuitive, as the nodes of the GAT model represent the delays, and the edges of the model represent the running, dwelling, and headway processes. Therefore, we used delays and the environment-related features as the node features, and the dwelling, running, and headway times as the edge features of their corresponding edge.

Let us take the graph representation in Fig. 8 as an example to illustrate the assignment of dwelling, running, and headway processes to the node feature in the second case (only node features). In Fig. 8, the features in terms of the train dwelling, running, and headway are assigned to the node where the connection originates. For example, for the dwelling process between node $E_{i,s}^a$ and node $E_{i,s}^d$, the planned and actual dwelling times are assigned to the node $E_{i,s}^a$, which is the origin of the dwelling process. Also, for the headway process between train i and train $i + 1$, the planned and actual arrival headways between train i and $i + 1$ at station s are assigned to the node $E_{i,s}^a$. The running process will be addressed in the same way the dwelling process is addressed. Intuitively, train delays and environmental factors (i.e., WS and RF) will also be assigned to the corresponding nodes.

Each model was repeated 10 times, and the mean value of the 10 runs was calculated and shown in Table 2. This table shows that addressing the factors as the node features can result in lower delay propagation errors. This is mainly because addressing all factors as node features can enable the model to take the planned running time (PRT) and planned headway (PH^d) from the current timeline to

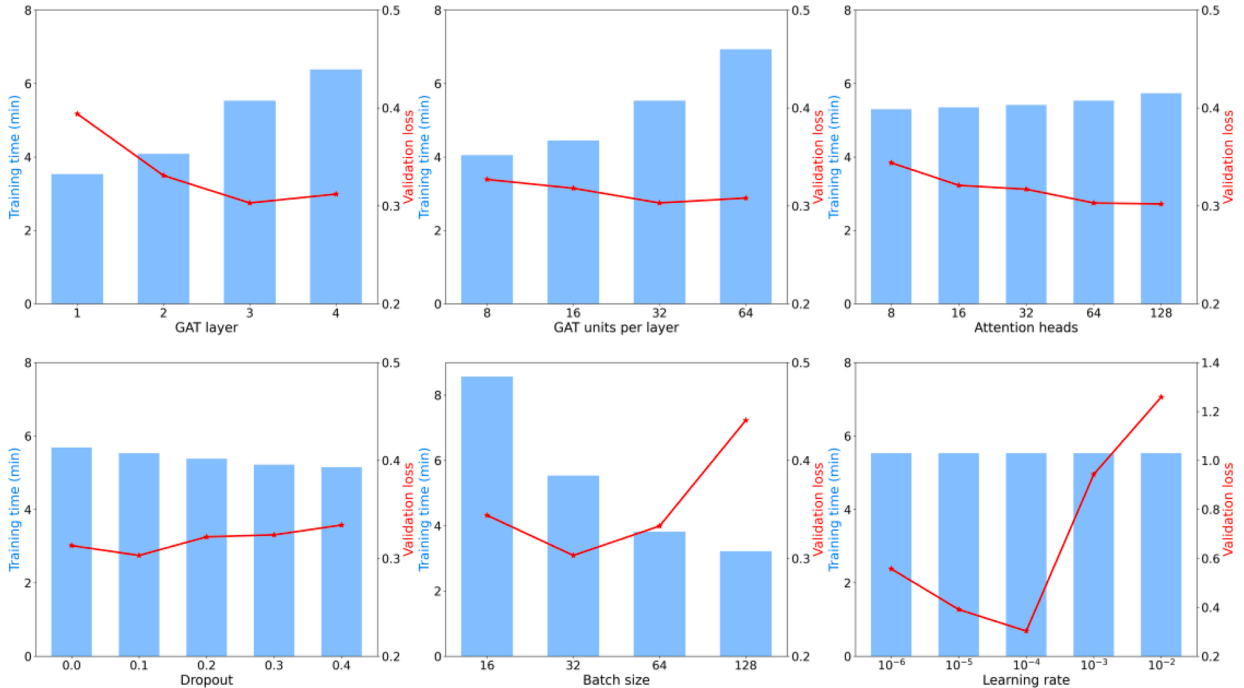


Fig. 7. The hyperparameter tuning results.

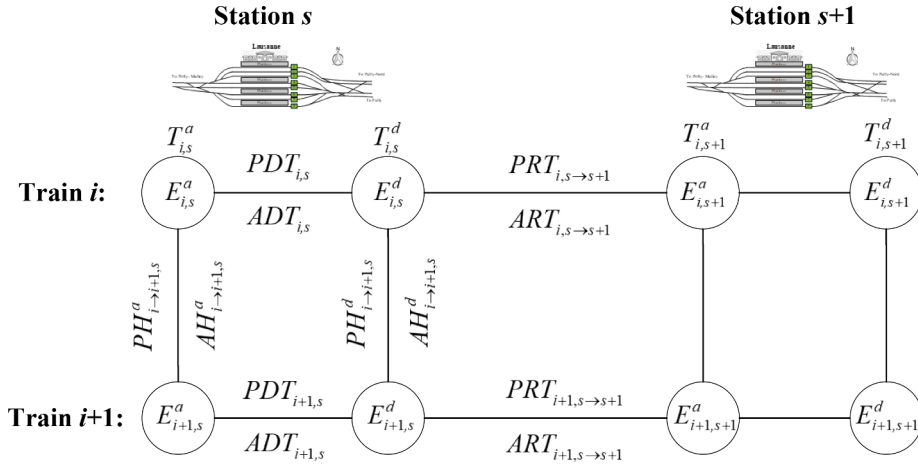


Fig. 8. An example of the assignment of timetable-related features in the GAT model.

the target (future) node as inputs. A graph regression task (i.e., predicting the target based on the whole graph) was performed in the present study, meaning that there are no arcs between the input (graph) and the output (target node). This means that the model addressing the process factors as edge features cannot include the factors PRT and PH^d from the current timeline to the target (future) node. However, these two factors are essential for predicting the target node (Li et al., 2021b; Nabian et al., 2019). Therefore, all the selected factors are treated as node features in the present study.

4.4. Experiments for train interactions

As introduced in Section 2, the existing train delay propagation models were built on a single train, or considering the interaction of two consecutive trains (shown in Fig. 1). To quantitatively show the number of trains that should be considered in the graph- or network-based models, we experimented on the GAT model considering one, two, three, and four trains. The structure of the delay propagation model considering two trains is shown in Fig. 4. However, we use the same paradigm in Fig. 4 to construct the delay propagation models when considering three and four trains. This means that the third and fourth trains are added to the graph similarly to adding the second train to the first train; see the comparison between Fig. 1(a) and (b). We also ran each experiment 10 times, and the mean value was calculated and shown in Table 3.

The results in Table 3 show that the graph model on two trains exhibits the lowest errors, meaning that considering two consecutive trains to establish the graph-based delay propagation model is appropriate. These results further prove the correctness of graph-based delay propagation models considering two consecutive trains (Corman and Kecman, 2018). This is probably caused by two factors combined. On the one hand, the train operation data was unbalanced, with both normal/punctual and delayed trains. Punctual trains account for approximately 90 %, based on statistics. On the other hand, the GAT model is based on messages passing between neighbor nodes. Unlike other machine learning and neural network models, the GAT model is built on graph structure. The topology of the graph is an essential factor to the model's performance. Training on more trains makes the model more likely to pass redundant information to the target node. For instance, when a normal train's delay is to be predicted, of which the preceding trains are delayed, the preceding train's information may lower the predictive performance of the prediction model on the to-be-predicted train; in this case, if two trains are considered in a graph, only one train's (the latter normal train) prediction results may be harmed; however, other punctual trains' prediction results can also be harmed if more trains are considered. However, because train interaction exists between consecutive trains, considering two trains is needed, as capturing train interaction is more beneficial than discarding the redundant information of the preceding train (see the comparison between considering one train and two trains in Table 3). Therefore, due to the message passing between nodes and graph structure, the GAT model is different from other traditional neural networks, e.g., the convolutional neural networks (Huang et al., 2021), which are more likely to perform better with more factors as inputs, because these neural networks learn near-zero weights for insignificant factors to avoid the influences of redundant features. However, for the graph-based model, the structure of the input data also counts, which can not be eliminated by learning insignificant weights.

Table 2
The performance of models under different feature assignment ways (min).

Experiment	MAE	RMSE
Only node features	0.303	0.690
Node features + edge features	0.813	1.253

Table 3

The performance of models considering the different numbers of trains (min).

Experiment	MAE	RMSE
One train	0.487	0.811
Two trains	0.303	0.690
Three trains	0.491	0.813
Four trains	0.642	0.958

4.5. Experiments for edge direction

In the GAT, edges between nodes can be unidirectional or bidirectional. In the traditional delay propagation models, e.g., the BN model, edges are unidirectional, meaning that the nodes are connected with time order. We use bidirectional graphs to establish the delay propagation model, which allows the model to address the overtaking problem in train operations. The structure of the proposed GAT model does not need to be changed when a train overtaking happens, compared to the existing graph-based delay propagation models, whose structure needs to be adjusted in real-time train traffic control (see discussion in (Corman and Kecman, 2018)). We experimented with the models using unidirectional and bidirectional graphs to show the significance of the proposed method. The GATs on unidirectional and bidirectional graphs had the same input and output; see Fig. 9 for details. The difference is that the unidirectional graph only allows messages to propagate in the forward direction, whereas bidirectional graphs allow messages to propagate in forward and backward directions.

We experimented on unidirectional and bidirectional graphs. Each type of graph was trained 10 times, and the mean value was calculated and shown in Table 4. Table 4 shows that the model on the bidirectional graph distinctively outperformed the model on the unidirectional graph. This is mainly because the traffic controllers not only considered the past information (e.g., delays), but estimated the future information (e.g., delays) to make a decision in practice. In other words, any traffic states (e.g., delays) are potentially influenced by their downstream and upstream states. The results in Table 4 further show the advantage of using a bidirectional graph for delay propagation. Because the proposed GAT model is performed based on the graph regression problem (similar to graph classification) (Veličković et al., 2017), where the inputs are the entire graph (details see Fig. 9), and the output is the future train delay, the bidirectional edges will not influence the ahead-of-prediction time of the model.

4.6. The proposed delay propagation model based on experiments

Once the feature assignment, train interaction, edge direction, and hyperparameters are optimized, the proposed delay propagation models can be determined using the graph structure in Fig. 4. Fig. 9 shows the model structure, the node features, and input and output. Nodes are numbered from 0 to 13 for simplicity. Based on the experiments in Section 4.3, the features are assigned to the corresponding nodes based on the feature assignment method in Fig. 8. In Fig. 9, the colored blocks above or under the nodes represent the features belonging to the corresponding node. The headway-related features were attributed to the preceding train (i.e., Node 0 to Node 6). Therefore, Node 0 has seven features, i.e., the T^a , PDT , ADT , PH^a , AH^a , WS , and RF ; Node 7 has five features, i.e., T^a , PDT , ADT , WS , and RF . Further, features in terms of the actual timetables can not be in the input of Nodes 6 and 12, which directly connect to the target node. Therefore, Node 6 has five features, i.e., without ADT and AH^a , compared to Node 0 to Node 5; Node 12 has four features, i.e., without ART compared to Node 7 to Node 11. Finally, the output node (Node 13) only has one feature, i.e., the target train delay (i.e., the arrival delay after the three considered stations).

Other than the feature assignment, other parameters set in the GAT model include: the GAT layers were followed by a fully-connected neural network layer with 128 neurons; the Adam optimizer (Kingma and Ba, 2014), was used to search the optima; a

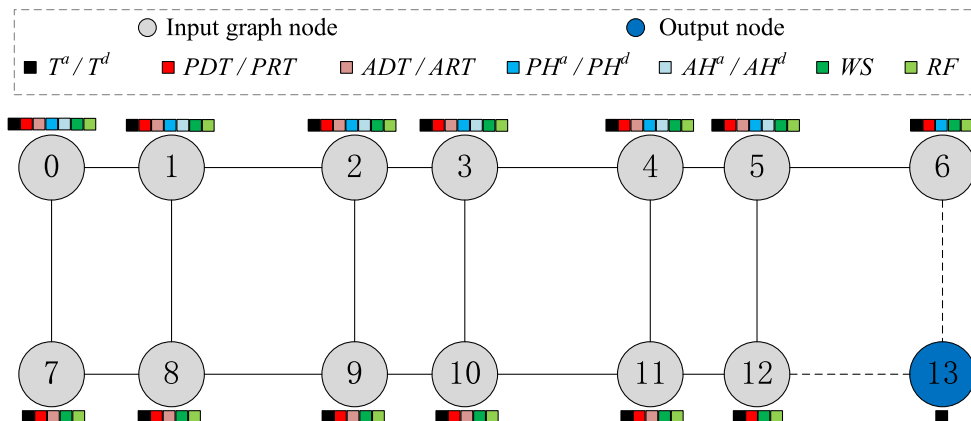
**Fig. 9.** The proposed delay propagation model in the GAT.

Table 4
The performance of models under different graphs (min).

Experiment	MAE	RMSE
Unidirectional graph	0.622	0.983
Bidirectional graph	0.303	0.690

standard loss function, mean absolute error, was chosen to measure the differences between the fitted and actual values; a self-loop is set on each input node, meaning each node influencing itself; the model was trained with 100 epochs.

5. Results

5.1. Attention coefficients for different edges

After training the model, the attention coefficient of each GAT layer can thus be obtained to show the importance of the message passing in the graph. Because 64 attention heads were used in each GAT layer, the mean values of the attention heads in each layer were calculated. We showed the attention coefficients of the train running process (the edges from departures to arrivals), dwelling process (the edge from arrival to departures), and headways (the connections between trains), in Fig. 10. Because three GAT layers were used in the present study, we separately show the distributions of the three layers. The edges in the model were separated into headway, dwelling, and running, and the mean values were calculated, respectively. We used the cumulative distribution function (CDF) to describe the distributions of the attention coefficients. Fig. 10 clearly shows that the distributions of attention coefficients for the train running process, dwelling process, and headways are distinctively different.

Generally, for each GAT layer, the attention coefficients of the dwelling process are the smallest, and those for the headways and running process are larger. This is understandable, as recovery times are mainly located in sections, and buffer times are always located between trains, making the edges for the train running process and headway more influential. However, train dwelling is generally a short process (e.g., over 90 % of the train dwelling processes cost less than 1 min), meaning that supplement/recovery times rarely exist at stations. This implies that the train dwelling process can hardly influence the delay propagation. Further, the train dwelling process is restricted to train departures. A typical example is that the trains can arrive earlier but cannot depart earlier. This means that the supplement/recovery times (if they exist) at stations are hardly used, which makes the train dwelling process less influential to the delay propagation. It should be noted that the above conclusions are based on the assumption that delay increases are much less frequent than delay recoveries in the delay propagation process; otherwise, the dwelling process should have been the most influential factor.

Overall, the distinctive differences in the attention coefficients for the running, dwelling, and headways imply that the connections between train events are diverse; the significance of the attention coefficients in the backward direction implies that the influence of factors on their previous (ahead-of-time) factors may also exist. These results further indicate the significance of using the GAT model to overcome the shortcomings (see details in Section 2) of previous delay propagation models.

Finally, to quantitatively show the differences between the attention coefficients, we performed a statistical test based on the common Kolmogorov-Smirnov (K-S) test (Lopes, 2011). The statistic of the K-S test is the distance (D) between the CDF curves of two observations (i.e., attention coefficients). The null hypothesis of this method is that the two sets of observations (i.e., the attention coefficients) follow the same distribution (we selected a significance level of 5 %). The testing results of the attention coefficients for the train running times, dwelling times, and headway times using the K-S method are shown in Table 5. This table shows that no results have a p-value more significant than 0.05, meaning that the null hypothesis is rejected. The testing results demonstrate that the attention coefficients in both forward and backward directions for the train running, dwelling processes, and headways are significantly different.

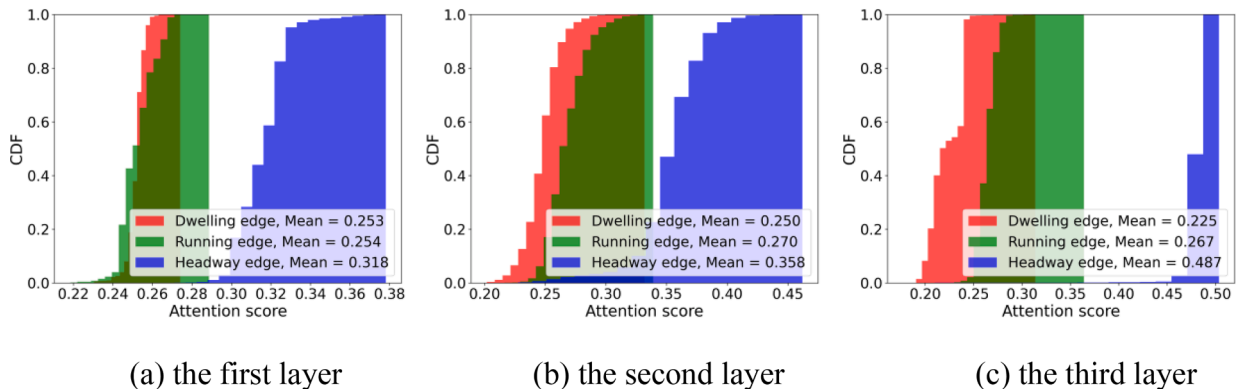


Fig. 10. The distribution of attention for the train running, dwelling, and headways in the (a)first, (b)second, and (c)third layer.

5.2. The determinant of attention scores

Section 5.1 shows that the attention coefficients for the running process, dwelling process, and headway are different. To identify the determinant of the attention coefficients, we performed a synergistic effect analysis between attention coefficients and the influence factors (i.e., running, dwelling, headways, and train delays). The analyses were performed based on the Spearman rank correlation coefficient because the relationships between the attention coefficients and influence factors may not be linear (see Fig. 11 as an example). Spearman rank correlation is a statistical measure of the strength and direction of monotonic correlation between variables, rather than a linear relationship (Hauke and Kossowski, 2011). It can deal with the non-linear relationships between two variables (V_1 and V_2). The Spearman rank correlation coefficient, which is based on rank variables, can be expressed by:

$$r_{(V_1, V_2)} = \frac{\text{cov}(R(V_1), R(V_2))}{\sigma_{R(V_1)} \sigma_{R(V_2)}} \quad (23)$$

Where $R(V_1)$ and $R(V_2)$ are the ranks (from small to large) of two variables V_1 and V_2 ; $\text{cov}(V_1, V_2)$ represents the covariance of variables V_1 and V_2 ; σ represent the standard deviation; $r_{(V_1, V_2)}$ represents the Spearman rank correlation of variables V_1 and V_2 . The greater absolute value of $r_{(V_1, V_2)}$ indicates stronger monotonicity.

Because the purpose is to find the determinant of the attention coefficient, we demonstrate the absolute value of the Spearman rank correlation coefficient ($r_{(V_1, V_2)}$), where the largest absolute value of $r_{(V_1, V_2)}$ represents the determinant. The $r_{(V_1, V_2)}$ between the learned attention coefficients and influence factors is shown in Fig. 12. It shows that the attention coefficients are noticeably proportional to headways, running/dwelling times, and delays. This means that these three factors have the most remarkable influence on delay propagation than others. They can be seen as the determinants of train delay propagation. The results are intuitive, as the recovery times are pre-scheduled in sections and stations, and buffer times are pre-scheduled between trains. Most importantly, this result is also consistent with that in Fig. 6, indicating the GAT model's ability to learn the delay propagation patterns.

To understand the relationship between the learned attention coefficients and the influential factors (i.e., delays, headways, and running/dwelling times), we showed scatter plots between them. Fig. 13 uses different colors to represent the learned attention coefficients. It shows that the attention coefficients are higher for longer headways, running/dwelling times, and delays. This means that the learned attention coefficients present positive relationships with these factors.

5.3. Quantitative explanations of learned attention scores

To explain train delay propagation, we first demonstrate a graph with learned attention scores. The learned attention scores in different directions are shown in different colors, with red fonts representing the learned attention scores from a node with a smaller number to a node with a larger one (e.g., from 0 to 1, from 1 to 8), greens representing the learned attention scores in the opposite direction, and blacks representing the learned attention scores of the self-loop of each node. It should be noticed that the learned attention scores are normalized, meaning that the summation of attention scores pointing to each node is 1. This indicates that each node can be fully explained by its neighbor nodes. From Fig. 14, we can observe that Node 9 has the most significant dependence on Node 8 (a previous node of the same train). The learned attention score from Node 8 to Node 9 is 0.39, meaning that the delay of Node 8 and the train running process from Node 8 to Node 9 have dominant influences on Node 9. However, Node 10 has the most significant dependence on Node 3 (the node of its preceding train). The learned attention score from Node 3 to Node 10 is 0.30, meaning that the delay of Node 3 and the headway times between Node 3 and Node 10 have dominant influences on Node 10. With the learned attention scores, traffic controllers can control train delay propagation more efficiently, as they know the determinant of delay propagation at each step/station. For instance, traffic controllers can speed up the train between Node 8 and Node 9 (from Asb to Ashd), and control the train running with the minimum headways between Node 3 and Node 10 (Train 93217 to Train 4035 at Ashd station) to reduce train delay propagation. The learned attention scores enable the traffic controllers to manage train delay propagation from a node perspective.

The proposed GAT model learns attention coefficients, representing the importance of one node's features to another. In the present study, each node contains multiple features. To interpret the importance of each feature to the learned attention coefficients, we performed regression analyses, where the learned attention coefficients were addressed as a response variable, and the node features were addressed as independent variables. This enables each learned attention score to be quantitatively explained by the features. To avoid multicollinearity, we used features in terms of delays, actual timetables, and environment in the regression analysis (i.e., T^a , T^d , ADT , ART , AH^a , AH^d , WS , and RF). Before performing regression, the independent factors were scaled to between zero and one, to avoid the influence of different dimensions. The contribution/explanation of each feature to the learned attention coefficients can be

Table 5
Consistency test results of attention coefficients for different connections.

Connections	First layer		Second layer		Third layer	
	Statistics	p-value	Statistics	p-value	Statistics	p-value
Running and dwelling times	0.269	< 2.2e-16	0.509	<2.2e-16	0.979	<2.2e-16
Running times and headways	0.998	< 2.2e-16	0.943	<2.2e-16	0.999	<2.2e-16
Dwelling times and headways	0.999	<2.20e-16	0.964	<2.2e-16	0.998	<2.2e-16

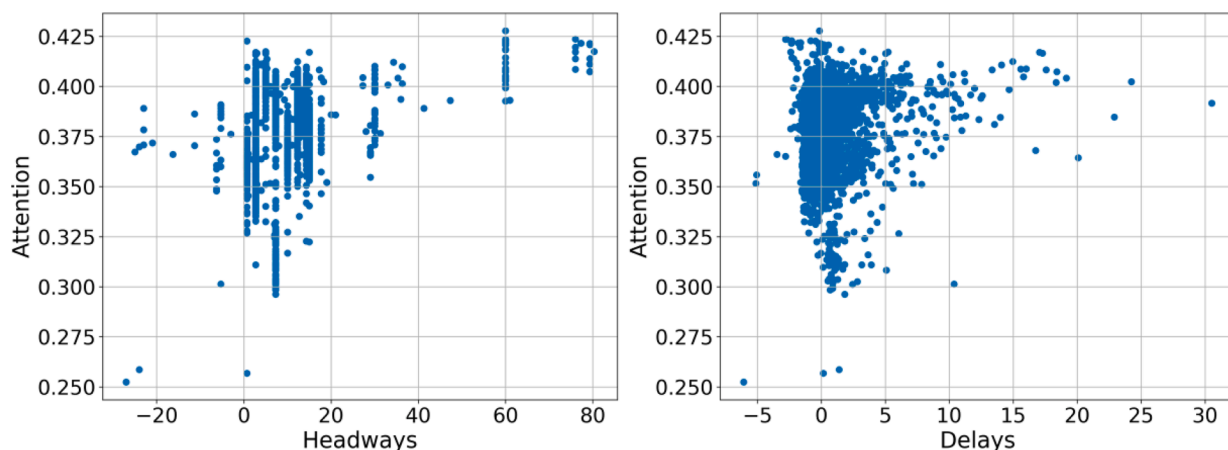


Fig. 11. The scatter plot between attention coefficients and influence factors (edge 3–10).

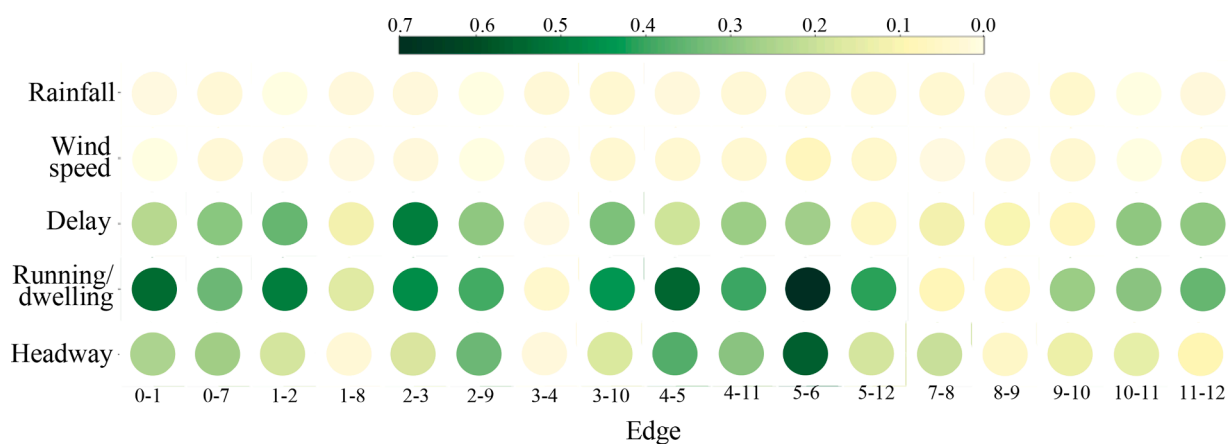


Fig. 12. Spearman rank correlation coefficients (the absolute value) between attention coefficients and influence factors.

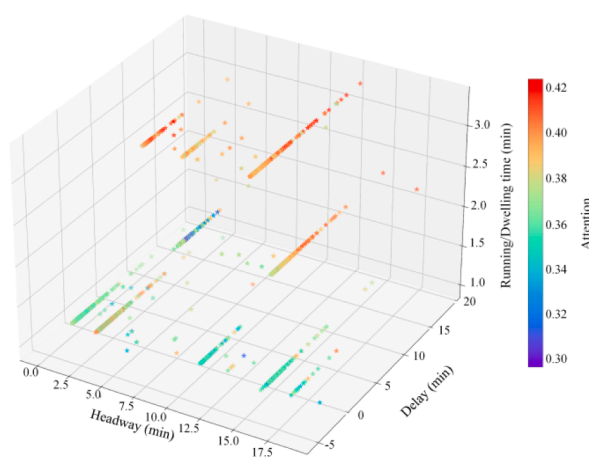


Fig. 13. The relationship between the attention coefficient, headway, delay, and running/dwelling process.

seen in Tables 6 to 8. In these tables, edges are demonstrated by two nodes (refer to Fig. 9); the regression coefficient of each factor is shown in the tables; the superscript with asterisks represents the significance (“***”, “**”, “*”, and “.” means the significance level of 0.001, 0.01, 0.05, and 0.1, respectively; no superscript means that the fitted coefficient is insignificant). With the regression

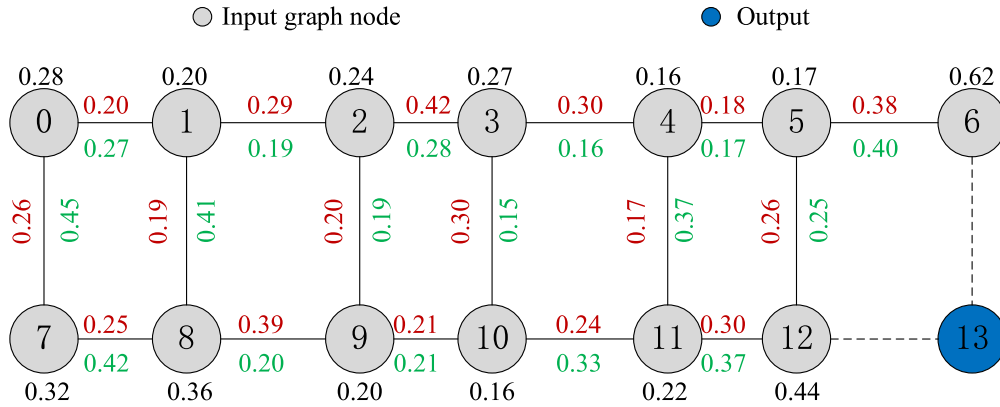


Fig. 14. The learned attention scores on a delay propagation graph.

coefficients, the quantitative influences of each factor on train delay propagation are clarified. For example, the regression results show that the actual dwelling time (ADT) greatly influences train delay propagation. This is intuitive, as dwelling times determine the recovery times for train delay recoveries.

Overall, the regression results in Tables 6 to 8 show that the train delays, headways, and running/dwelling times influence the attention coefficients more significantly than the environment-related factors. For the headway connections (the green rows), the significant factors (delays, headways, and running/dwelling) tend to influence the attention coefficients positively. However, the significant factors negatively affect the attention coefficient of the dwelling and running edges. In addition, the headway times are more influential on the attention coefficients of the headway edges. This is because longer headways (i.e., more buffer times between trains) are more likely to absorb train delays, leading to smaller attention coefficients for dwelling and running processes in Fig. 10 (but larger attention coefficients for headways). It can be explained by the principle that the recovery times only account for 3 %-7% of the minimum (required) travel times in the timetable for the European railways (Hansen and Pachl, 2014). The fitted regression results in Tables 6 to 8 clarify the contribution of each feature to the learned attention scores, meaning that traffic controllers can manage train delay propagation from a feature perspective.

5.4. Predictive performance

(1) Residual analyses

Sections 5.1 and 5.2 demonstrate that headway edges present the most significant attention coefficients, and headways, delays, and running/dwelling times are the most influential factors. Therefore, we first analyzed the distribution of residuals over these three influential factors, shown in Fig. 15. In this figure, the red lines are the regression lines based on the scatters. This figure clearly shows that the proposed model's performance will not worsen with the increase in headways and running times. This indicates that the proposed model is unbiased on headways and running times. However, the model's predictive errors grow slightly with the increase in delays. This is because the proposed model predicts train delays (the label/output is train delays). Long delays have lower frequencies, meaning fewer cases to train the model. This inevitably led to lower predictive performance of the model. This result has also been proven by previous train delay prediction studies (Corman and Kecman, 2018; Huang et al., 2020b). This result also confirms our previous analysis in Section 4.1 that data imbalance may lead to a biased model.

(2) Comparative analyses

Delay propagation models can usually be used for delay prediction. To demonstrate the predictive performance of the proposed delay propagation model, we also compared it with the other two state-of-the-art graph- or network-based delay propagation models. One is the BN, which has been extensively used for train delay propagation and other train delay-related problems (Corman and Kecman, 2018; Huang et al., 2020a; Lessan et al., 2019). The other is the simple graph convolutional networks (GCN), i.e., the GCN without edge weights, which recently have been used for train delay propagation in railway networks (Zhang et al., 2021; Zhang et al., 2022). Another baseline model is a graph model with the minimum process times (e.g., dwelling times, running times, and headways)

Table 6

Explanations (fitted regression coefficients) to the learned attention coefficients of dwelling processes.

Edge	Intercept	RF	WS	T^a	ADT	AH^a/AH^d	R^2
0-1	0.155***	-0.001	0.000	-0.024***	-0.070***	-0.008	0.322
2-3	0.144***	-0.004*	0.003*	-0.066***	-0.021***	0.002	0.334
4-5	0.144***	-0.006***	0.005***	-0.008	-0.018***	-0.030***	0.276
7-8	0.345***	0.002	-0.001	0.026***	0.005***	-0.022***	0.038
9-10	0.297***	-0.002	0.004*	-0.004	0.010***	0.006	0.059
11-12	0.372***	0.003*	-0.005**	-0.037***	-0.012***	-0.017***	0.165

Table 7

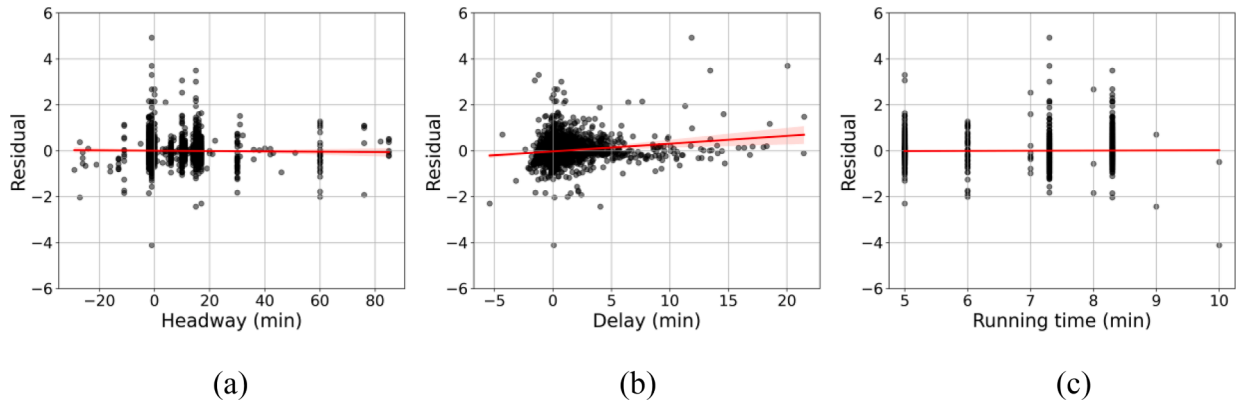
Explanations (fitted regression coefficients) to the learned attention coefficients of running processes.

Edge	Intercept	RF	WS	T^d	ART	AH^a/AH^d	R^2
1–2	0.149***	0.000	0.002	−0.071***	−0.024***	−0.017***	0.137
3–4	0.132***	−0.001	0.002	−0.022***	0.002	−0.026***	0.030
5–6	0.411***	0.001	−0.006***	0.004	−0.039***	−0.016***	0.560
8–9	0.327***	0.002	−0.004*	−0.039***	0.008***	0.008*	0.027
10–11	0.299***	−0.001	−0.001	0.030***	0.022***	−0.008	0.152

Table 8

Explanations (fitted regression coefficients) to the learned attention coefficients of headway connections.

Edge	Intercept	RF	WS	T^a/T^d	ADT/ART	AH^a/AH^d	R^2
0–7	0.416***	−0.001	0.000	0.040***	0.016***	0.032***	0.154
1–8	0.384***	0.001	−0.001	0.023***	0.019***	0.006	0.071
2–9	0.378***	0.003	−0.003	0.040***	0.009***	0.070***	0.207
3–10	0.377***	0.004*	0.000	0.034***	0.026***	0.018***	0.218
4–11	0.376***	0.004*	0.000	0.046***	0.017***	0.025***	0.190
5–12	0.394***	0.002	−0.004**	0.029***	−0.046***	0.072***	0.316

**Fig. 15.** The residual distribution over the (a)headway, (b)delay, and (c)running time.

as the edge weights, proposed in (Goverde, 2010), called Graph-MPT. In the present study, we used the minimum dwelling times, running times, and headways in the observations (after normalization) as the edge weights. The last baseline model is derived from the literature (Kecman and Goverde, 2014), called Graph-RC, where the edge weights of the graph model are the regression coefficients of linear models. The authors addressed the dwelling and running process dependent on the arrival and departure events in the study. Therefore, they used arrival and departure delays to explain the dwelling and running variations, obtaining the regression coefficients as the edge weights. They used the minimal headways from the observation as the representative values for the connection between trains. Therefore, we also used the regression coefficients and the minimal headways based on the observations as the edge weights of the baseline model Graph-RC. For the two graph models with minimum process times and regression coefficients as edge weights, we implement the graph based on the GCN model, and add the edge weights (obtained separately) to the GCN graph.

For a fair comparison, we also tuned the parameters of the benchmark models. The BN model has no parameter that needs to be tuned, because building on different numbers of trains does not influence its performance due to its Markov property. Because the GAT and the other three benchmark models are variants of the GCN, the parameters optimized for the GAT model were considered, including the number of layers, units per layer, dropout rate, batch size, learning rate, feature assignment, train interaction, and edge direction. The benchmark models' parameters can be selected similarly to the proposed GAT model. Therefore, the alternative parameter values in Sections 4.2 to 4.5 were used in the parameter-tuning process of the benchmark models. Because Graph-MPT and Graph-RC are obtained by manually adding edge weights to the GCN, we only tuned the parameters based on the GCN model. The optimal parameters were directly used in Graph-MPT and Graph-RC models. Based on experimental results, two layers, each with 32 units, dropout rate = 0.3, batch size = 8, and learning rate = 0.0001, were selected for the GCN model. Node features, two trains, and bidirectional edges were used in the benchmark models. The benchmark models and the proposed model all performed better on node features, two trains, and bidirectional graphs. This is mainly because they are all variants of graph neural networks, exhibiting similar properties on train operation data.

The BN model is chosen as the benchmark to show the superiority of the GAT model over the standard delay propagation models in the existing studies. The GCN model is the base of the GAT model (the GCN is a GAT without the attention mechanism). Therefore, the

GCN model is selected as the benchmark to show the performance improvements of the GAT model over the GCN model. The performance improvement of the GAT lies in the attention mechanism, which is the motivation of the present study. The GCN models with minimum process times and regression coefficients as edge weights are selected as the baseline models to show the performance of the proposed GAT model compared against the graph models with edge weights proposed in the existing studies (Goverde, 2010; Kecman and Goverde, 2014). Each model was run 10 times, and the mean value of the 10 runs was calculated and demonstrated.

We first compared the residual distributions of the GAT and the benchmark models, as shown in Fig. 16(a). This figure indicates that the residual distribution curve of the GAT model (the black curve), located on the top-left side of the figure, represents smaller predictive errors. In addition, Fig. 16(b) demonstrates the distribution of the absolute residuals, which exhibits that the box body of the BN model is the highest, whereas that of the GAT model is the lowest. This indicates that the GAT model has the lowest errors. It is worth noticing that the GAT and BN models exhibit much fewer outliers (worse cases) than the GCN, Graph-MPT, and Graph-RC, on the top of the box, indicating that the GAT and BN models are superior to these models on the extreme cases (e.g., the cases with a big delay jump between two stations). However, compared against the BN model, the GAT model still exhibits better performance, as (1) the outliers of residuals of the GAT model are slightly fewer than the BN model, (2) the GAT model has significantly lower mean errors than the BN model.

The proposed GAT model is a point-estimation model, i.e., predicting a deterministic value for each case. The model can provide the traffic controllers with estimated values of future train delays for decision-making. To systematically evaluate a point-estimation model, we used two mean-value-based metrics, i.e., MAE and RMSE, a relative metric, i.e., mean absolute percentage error (MAPE), and a median-based metric, i.e., the median absolute error (MedAE). The MedAE metric represents the median lines in the boxes in Fig. 16(b). We used these metrics to evaluate the models' performance on the whole dataset and the delayed trains, i.e., delays longer than 3 min (Goverde, 2005).

Tables 9 and 10 demonstrate that the GAT model considerably outperforms the benchmark models. The results in Fig. 16 and Tables 9 and 10 demonstrate the model's ability to discern the factor importance, enabling it to surpass other standard delay propagation models. The results further imply the significance of the attention mechanism for the train delay propagation (i.e., context-aware delay propagations).

Finally, we also investigated the importance of features to the predictive performance of the model. The proposed GAT model was trained on all the features shown in Section 3.2. To that end, we removed each feature set from the dataset, and trained the GAT model on the new datasets to demonstrate its sensitivity to features. Because previous train delays are obviously crucial to future delays, we only experimented with other timetable- and environment-related features. We removed FPT, FAT, WS, and RF (see Section 3.2) from the data, and ran the GAT model on each new dataset 10 times. The distributions of MAE and RMSE of the model on the new data are shown in Fig. 17.

Fig. 17 shows that removing any feature harmed the predictive performance of the model. It shows that the FAT is the most essential feature among them. The WS and RF had minor importance on the model compared to FAT and FPT, but removing them still harmed the model's accuracy and stability.

5.5. Practicability evaluation

To evaluate the practicability of the models, we show the model's training time (computational cost) and response time. All the models can be well-trained and saved in advance. Here, we assume they will be loaded to memory and used in real-time decision-making. Therefore, the response time of the models includes the model loading and prediction executing time. The results are shown in Table 11.

We can see from Table 11 that the BN model can be quickly trained and executed. However, a new BN structure will be needed once

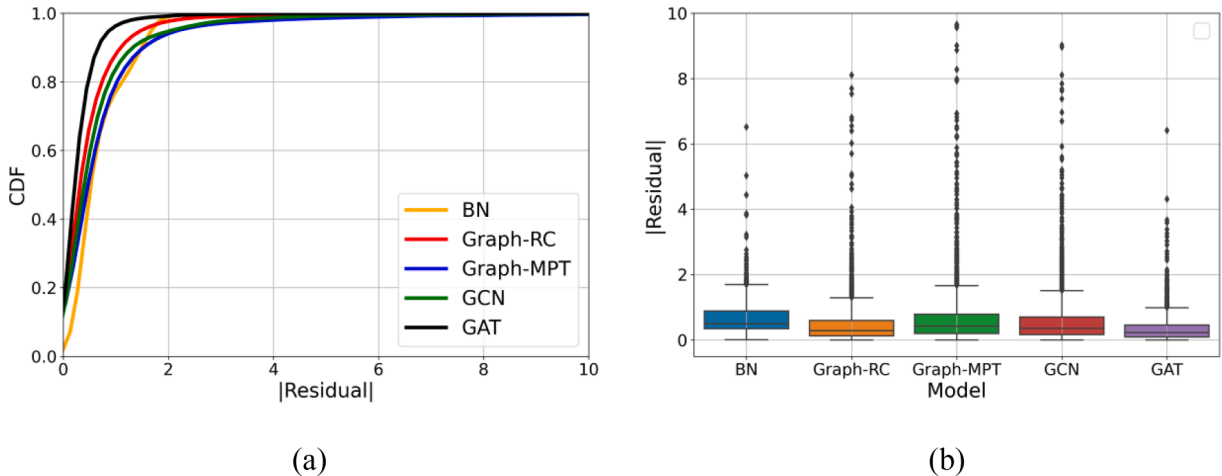


Fig. 16. The (a)CDF and (b)boxplot of residuals of different models.

Table 9

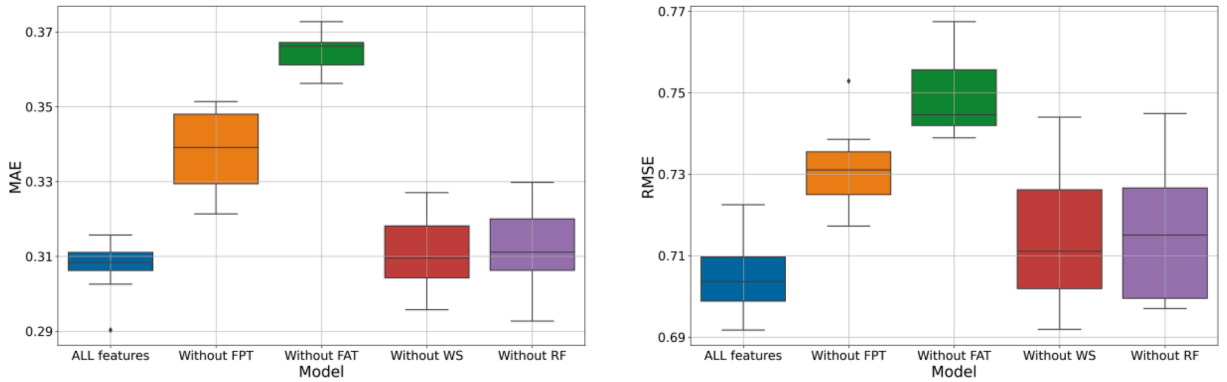
Predictive errors of delay propagation models on the whole testing samples.

Model	MAE (min)	RMSE (min)	MedAE
BN	0.692	0.990	0.494
GCN	0.557	1.132	0.359
Graph-MPT	0.681	1.393	0.425
Graph-RC	0.432	0.878	0.287
GAT	0.303	0.690	0.195

Table 10

Predictive errors of delay propagation models on the samples with actual delay longer than 3 min.

Model	MAE (min)	RMSE (min)	MAPE	MedAE
BN	1.174	2.726	11.8 %	0.823
GCN	2.143	3.170	32.2 %	1.117
Graph-MPT	1.894	3.114	27.1 %	1.424
Graph-RC	1.443	2.464	19.6 %	1.451
GAT	0.667	2.563	5.8 %	0.214

**Fig. 17.** The sensitivity of features to the predictive performance of the model.

a train overtaking happens because the BN model is unidirectional. This considerably lowers the convenience of the BN model in real-time decision-making. Further, many models with diverse structures need to be constructed and trained before covering all possible train overtaking situations, e.g., train overtaking at different stations.

The GCN, Graph-MPT, Graph-RC, and GAT models are neural network models, built on bidirectional graphs. First, the influences of the attention mechanism on the model's efficiency can be clarified by comparing the times of the GCN and GAT in Table 11. The comparison shows that the GAT model only has much longer model-loading times. This is primarily because the GAT model needs to load the parameters for attention calculation. Generally, these models have similar training times, much longer than the BN model. Although training the GAT model costs approximately 5.6 min, and the GAT model is the slowest one among the models in loading and predicting, it is fast enough for real-time decision-making (only 0.187 and 0.007 seconds needed).

To systematically evaluate the model's practicability, we demonstrate the influences of data volumes on the model's performance in terms of accuracy and efficiency. We randomly selected 30 % of the data as testing data, and the remaining data were used for model parameter learning and model validation. Therefore, we investigated the model's accuracy and efficiency based on the remaining (70 %) data. The remaining data (70 %) were always randomly separated into two datasets, including 10 % and 90 %, 20 % and 80 %, 30 % and 70 %, 40 % and 60 %, 50 % and 50 %, 60 % and 40 %, 70 % and 30 %, 80 % and 20 %, 90 % and 10 %, one part for learning the model's parameters, and another for validation of the model. We ran the model on different data volumes 10 times, and the average values were calculated. Fig. 18 demonstrates the model's accuracy (regarding MAE and RMSE) on the testing dataset and the time cost of model training on different data volumes. We can conclude from this figure that data volume significantly influenced the model's performance. The figure shows that neither a large percentage of training nor validation data resulted in the best model. Considering the predictive performance, 80 % and 20 % for parameter learning and validation resulted in the best performance. However, 20 % and 80 % for parameter learning and validation resulted in the best performance regarding the model's training efficiency. The total data volume was constant, resulting in dependencies between the learning and validation processes. For example, if only 10 % of the data were used for parameter training and the remaining 90 % for validation, the parameter learning process would take a short time; however, the validation process would take a long time. Because the GAT model can be learned in advance, we chose the best model regarding predictive errors, meaning that the data were separated into 80 % and 20 % for parameter learning and validation,

Table 11

The time cost of models.

Model	Training time (min)	Response time (s)	
		Model loading per time	Prediction per sample
BN	0.3	0.008	<0.001
GCN	4.3	0.017	0.004
Graph-MPT	4.5	0.021	0.005
Graph-RC	4.5	0.022	0.005
GAT	5.6	0.187	0.007

Note: All the experiments in the study were conducted on a device with an AMD R7 2700 CPU and an RTX 2070 super GPU. The BN model was computed by the CPU, whereas other graph models were computed by the GPU.

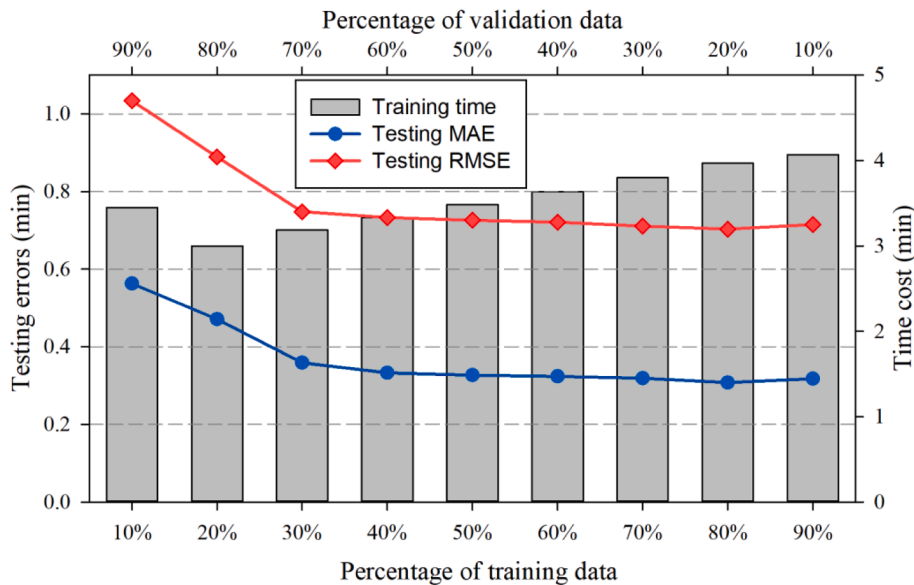


Fig. 18. The influence of data volumes on the model's performance. (All the experiments were conducted on a device with an AMD R7 2700 CPU and an RTX 2070 super GPU.).

respectively, in the present study.

6. Conclusions and future directions

We put forward a delay propagation model based on a graph neural network model, i.e., the GAT model. The GAT model is built on a graph, enabling it to retain the high interpretability of the traditional graph/network models for delay propagation (i.e., using nodes to represent train events, instead of treating train operations as a black box). In addition, the proposed GAT model uses an attention mechanism to capture the strength-of-influence of factors to delay propagation, and can assign various weights to edges between factors. This quality allows the GAT model to pay more attention (represented by the learned attention coefficients) to the significant factors, and be explainable to train operation and delay propagation. In comparison, the existing delay propagation models address the influences of factors as constant, which are unaware of the contributions of each factor. Further, the proposed model is built on bidirectional graphs, allowing the model to overcome the weakness of the traditional delay propagation models whose structure needs to be updated when a train overtaking happens. We tested the proposed model on the dataset from the Dutch railway line and gave detailed analyses and interpretations of the learned attention coefficients. The results show that: (1) the attention coefficient obtained by the GAT model for the train running process, dwelling process, and headways are distinctively different, further showing the significance of addressing the connections between train events with different weights; (2) the headway which has the most significant influences, is the determinant of train delay propagation; (3) the predictive performance of the model on train delay prediction problem is considerably superior to a standard network-based train delay propagation model, and a state-of-the-art graph-based delay prediction model.

The advantages of the proposed method lie in ensuring adaptive edge weights in the graph, explaining delay propagation from a micro perspective, and avoiding structure changes in real-time train delay prediction under the situation of a train overtaking. The GAT model uses multi-head attention to learn edge weights from data, enabling the model to learn appropriate edge weights based on the interactions between nodes. The multi-head can uncover different correlations between factors (e.g., linear, nonlinear). In comparison,

manually assigning edge weights usually can only capture one correlation, easily leading to inaccurate information (e.g., delays) propagation between nodes. The explanation of the GAT model is mainly reflected by the learned attention scores, which can explain the train delay propagation from two aspects: (1) the learned attention scores represent the influence of neighbor nodes on the target node, which clearly shows the contributions of each train event to the future train event, i.e., future train delay; (2) with the learned attention scores, the importance of each factor on the future train delays can also be quantified by using a regression method. With these two aspects, the GAT model enables traffic controllers to better understand the train delay propagation, e.g., the contribution of each train event, the determinant factor to future train delays.

The GAT model can also be extended to other problems in terms of train operation management. First, because the GAT model is able to learn edge weights from data. The strengths of connections between nodes can thus be quantified. It can be used to learn more appropriate model structures considering arbitrary trains, as graph structures in this and previous studies were all based on fixed paradigms, see details in (Corman and Kecman, 2018; Lessan et al., 2019). Further, using the learned edge weights to represent the processing times of trains (i.e., running and dwelling times), the GAT model is expected to be used in retiming the train operations under interruption situations. In this situation, the performance of the GAT model (used in train retiming) will not outperform the best case in the dataset, because the edge weights are learned from historical data. Finally, the proposed method can be extended to a network-wide delay propagation problem. This is because the graph structure can be kept constant, and modifications are only needed in the dataset. We need to ensure that every training sample always includes the features of consecutive trains because a train may have diverse consecutive trains on different lines. For instance, a train runs from Amsterdam to Eindhoven through Utrecht station (see Fig. 5 for the train path). If the delays of this train at Asb and Hdl are predicted, the preceding train of the target train on the A-U and U-E lines, respectively, shall be used to build the delay propagation graph.

This study is imperfect, arising from the GAT model and how the delay propagation model is formulated. From the GAT model's perspective, the edge weights of the GAT model are computed from a node level based on data, making the model incapable of obtaining the importance coefficients of each node feature. This means that the importance of features can only be computed based on the learned edge weights by using other methods (e.g., regression). Further, because the edge weights are learned from data, they are easily affected by the data distribution. For example, the model may be under-fitting on the low-frequency cases, making the edge weights less accurate on long delays (longer delays have lower frequencies). From the model formulation perspective, we built the model purely based on expert knowledge (i.e., the graph's structure). In reality, some links/connections may not be needed in the structure, or more trains may need to be considered by removing some insignificant connections. In future studies, we will aim to overcome these weaknesses. For the model, some sampling techniques, e.g., up sampling and down sampling, can be used to mitigate the underfitting of the GAT model on low-frequency cases. Additionally, for more appropriate delay propagation representations, we will explore more flexible model structures built on arbitrary trains, by removing some insignificant connections, to improve the model structure's rationality.

CRedit authorship contribution statement

Ping Huang: Conceptualization, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing, Funding acquisition. **Jingwei Guo:** Validation, Writing – review & editing. **Shu Liu:** Conceptualization, Methodology, Software, Validation, Writing – original draft. **Francesco Corman:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the National Nature Science Foundation of China [grant numbers 72301221] and the Swiss National Science Foundation, Switzerland, under the research project DADA [grant numbers 181210].

References

- Artan, M.Ş., Şahin, İ., 2021. Exploring patterns of train delay evolution and timetable robustness. *IEEE Trans. Intell. Transp. Syst.* 23 (8), 11205–11214.
- Barta, J., Rizzoli, A.E., Salani, M., Gambardella, L.M., 2012. Statistical modelling of delays in a rail freight transportation network, *Proceedings of the 2012 Winter Simulation Conference (WSC)*. IEEE, pp. 1–12.
- Berger, A., Gebhardt, A., Müller-Hannemann, M., Ostrowski, M., 2011. Stochastic delay prediction in large train networks, 11th Workshop on Algorithmic Approaches for Transportation Modelling, Optimization, and Systems. Schloss Dagstuhl-Leibniz-Zentrum fuer Informatik.
- Binder, S., Maknoon, Y., Bierlaire, M., 2017. The multi-objective railway timetable rescheduling problem. *Transp. Res. Part C: Emerging Technol.* 78, 78–94.
- Büker, T., Seybold, B., 2012. Stochastic modelling of delay propagation in large networks. *J. Rail Transp. Plann. Manage.* 2 (1–2), 34–50.

- Carey, M., Kwieciński, A., 1994. Stochastic approximation to the effects of headways on knock-on delays of trains. *Transp. Res. B: Methodol.* 28 (4), 251–267.
- Corman, F., Kępcan, P., 2018. Stochastic prediction of train delays in real-time using Bayesian networks. *Transp. Res. Part C: Emerging Technol.* 95, 599–615.
- Corman, F., Meng, L., 2014. A review of online dynamic models and algorithms for railway traffic management. *IEEE Trans. Intell. Transp. Syst.* 16 (3), 1274–1284.
- Feng, T., Lusby, R.M., Zhang, Y., Peng, Q., Shang, P., Tao, S., 2023. An ADMM-based dual decomposition mechanism for integrating crew scheduling and rostering in an urban rail transit line. *Transp. Res. C: Emerging Technol.* 149, 104081. <https://doi.org/10.1016/j.trc.2023.104081>.
- Fischetti, M., Salvagnin, D., Zanette, A., 2009. Fast approaches to improve the robustness of a railway timetable. *Transp. Sci.* 43 (3), 321–335.
- Gaurav, R., Srivastava, B., 2018. Estimating Train Delays in a Large Rail Network Using a Zero Shot Markov Model, 2018 21st International Conference on Intelligent Transportation Systems (ITSC). IEEE, pp. 1221–1226.
- Ge, X., Jiang, C., Qin, Y., Huang, P., 2023. Modelling the cascading effects of train delay patterns and inter-train control actions with Bayesian networks. *Int. J. Rail Transp.* 1–26.
- Goverde, R.M., 2005. Punctuality of railway operations and timetable stability analysis. *Trail, The Netherlands*.
- Goverde, R.M., 2010. A delay propagation algorithm for large-scale railway traffic networks. *Transp. Res. Part C: Emerging Technol.* 18 (3), 269–287.
- Hansen, I.A., Goverde, R.M., van der Meer, D.J., 2010. Online train delay recognition and running time prediction, Intelligent Transportation Systems (ITSC), 2010 13th International IEEE Conference on. IEEE, pp. 1783–1788.
- Hansen, I., Pachl, J., 2014. Railway timetabling and operations: Analysis, modelling, optimisation, simulation, performance, evaluation. Eurail press.
- Hauke, J., Kossowski, T., 2011. Comparison of values of Pearson's and Spearman's correlation coefficients on the same sets of data. *Quaestiones Geographicae* 30 (2), 87.
- Heglund, J.S., Taleongpong, P., Hu, S., Tran, H.T., 2020. Railway Delay Prediction with Spatial-Temporal Graph Convolutional Networks, 2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC). IEEE, pp. 1–6.
- Huang, P., Lessan, J., Wen, C., Peng, Q., Fu, L., Li, L., Xu, X., 2020a. A Bayesian network model to predict the effects of interruptions on train operations. *Transp. Res. Part C: Emerging Technol.* 114, 338–358.
- Huang, P., Wen, C., Fu, L., Lessan, J., Jiang, C., Peng, Q., Xu, X., 2020b. Modeling train operation as sequences: A study of delay prediction with operation and weather data. *Transp. Res. Part E: Logistics and Transp. Rev.* 141, 102022.
- Huang, P., Wen, C., Fu, L., Peng, Q., Tang, Y., 2020c. A deep learning approach for multi-attribute data: A study of train delay prediction in railway systems. *Inf. Sci.* 516, 234–253.
- Huang, P., Li, Z., Wen, C., Lessan, J., Corman, F., Fu, L., 2021. Modeling train timetables as images: A cost-sensitive deep learning framework for delay propagation pattern recognition. *Expert Syst. Appl.* 177, 114996.
- Huang, P., Li, Z., Zhu, Y., Wen, C., Corman, F., 2023. Train traffic control in merging stations: A data-driven approach. *Transp. Res. Part C: Emerging Technol.* 152, 104155.
- Huang, P., Spanniger, T., Corman, F., 2022. Enhancing the Understanding of Train Delays With Delay Evolution Pattern Discovery: A Clustering and Bayesian Network Approach. *IEEE Trans. Intell. Transp. Syst.* <https://doi.org/10.1109/ITITS.2022.3140386>.
- Jiang, Z., Xu, R., Xie, C., 2007. Train delay propagation simulation in rail transit system. *International Conference on Transportation Engineering 2007*, 789–794.
- Kępcan, P., Corman, F., Meng, L., 2015. Train delay evolution as a stochastic process, 6th International Conference on Railway Operations Modelling and Analysis-RailTokyo2015.
- Kępcan, P., Goverde, R.M., 2014. Online data-driven adaptive prediction of train event times. *IEEE Trans. Intell. Transp. Syst.* 16 (1), 465–474.
- Kępcan, P., Goverde, R.M., 2015. Predictive modelling of running and dwell times in railway traffic. *Public Transport* 7 (3), 295–319.
- Keiji, K., Naohiko, H., Shigeru, M., 2015. Simulation analysis of train operation to recover knock-on delay under high-frequency intervals. *Case Studies on Transport Policy* 3 (1), 92–98.
- Kingma, D.P., Ba, J., 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*.
- Lessan, J., Fu, L., Wen, C., 2019. A hybrid Bayesian network model for predicting delays in train operations. *Comput. Ind. Eng.* 127, 1214–1222.
- Li, B., Guo, T., Li, R., Wang, Y., Ou, Y., Chen, F., 2021a. Delay Propagation in Large Railway Networks with Data-Driven Bayesian Modeling. *Transp. Res. Rec.* 2675 (11), 472–485.
- Li, Z., Wen, C., Hu, R., Xu, C., Huang, P., Jiang, X., 2021b. Near-term train delay prediction in the Dutch railways network. *Int. J. Rail Transp.* 9 (6), 520–539.
- Li, Z., Huang, P., Wen, C., Jiang, X., Rodrigues, F., 2022. Prediction of train arrival delays considering route conflicts at multi-line stations. *Transp. Res. Part C: Emerging Technol.* 138, 103606.
- Lopes, R.H.C., 2011. Kolmogorov-Smirnov Test. In: Lovric, M. (Ed.), *International Encyclopedia of Statistical Science*. Springer, Berlin Heidelberg, Berlin, Heidelberg, pp. 718–720.
- Meester, L.E., Muns, S., 2007. Stochastic delay propagation in railway networks and phase-type distributions. *Transp. Res. Part B: Methodol.* 41 (2), 218–230.
- Meloni, C., Pranzo, M., Samà, M., 2021. Risk of delay evaluation in real-time train scheduling with uncertain dwell times. *Transp. Res. Part E: Logistics Transp. Rev.* 152, 102366.
- Milinković, S., Marković, M., Vesković, S., Ivić, M., Pavlović, N., 2013. A fuzzy Petri net model to estimate train delays. *Simul. Model. Pract. Theory* 33, 144–157.
- Nabian, M.A., Alemazkoor, N., Meidani, H., 2019. Predicting near-term train schedule performance and delay using bi-level random forests. *Transp. Res. Rec.* 2673 (5), 564–573.
- Nash, A., Huerlimann, D., 2004. Railroad simulation using OpenTrack. *WIT Transactions on the Built Environment* 74.
- Şahin, İ., 2017. Markov chain model for delay distribution in train schedules: Assessing the effectiveness of time allowances. *J. Rail Transp. Plann. Manage.* 7 (3), 101–113.
- Shi, R., Xu, X., Li, J., Li, Y., 2021. Prediction and analysis of train arrival delay based on XGBoost and Bayesian optimization. *Appl. Soft Comput.* 109, 107538.
- Spanniger, T., Trivella, A., Büchel, B., Corman, F., 2022. A review of train delay prediction approaches. *J. Rail Transp. Plann. Manage.* 22, 100312.
- Ulak, M.B., Yazici, A., Zhang, Y., 2020. Analyzing network-wide patterns of rail transit delays using Bayesian network learning. *Transp. Res. Part C: Emerging Technol.* 119, 102749.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I., 2017. Attention is all you need. *Advances in neural information processing systems* 30.
- Veličković, P., Cucurull, G., Casanova, A., Romero, A., Lio, P., Bengio, Y., 2017. Graph attention networks. *arXiv preprint arXiv:1710.10903*.
- Wang, P., Goverde, R.M., 2017. Multi-train trajectory optimization for energy efficiency and delay recovery on single-track railway lines. *Transp. Res. Part B: Methodol.* 105, 340–361.
- Wang, R., Work, D.B., 2015. In: *Data Driven Approaches for Passenger Train Delay Estimation*, pp. 535–540.
- Wen, C., Huang, P., Li, J., Peng, Q., Jiang, C., 2016. Identifying key trains and their reasonable ratios in high-speed rail timetables. *J. Transp. Saf. & Sec.* 8 (supl), 118–135.
- Wen, C., Li, Z., Lessan, J., Fu, L., Huang, P., Jiang, C., 2017. Statistical investigation on train primary delay based on real records: evidence from Wuhan-Guangzhou HSR. *Int. J. Rail Transp.* 5 (3), 170–189.
- Wen, C., Huang, P., Li, Z., Lessan, J., Fu, L., Jiang, C., Xu, X., 2019. Train dispatching management with data-driven approaches: a comprehensive review and appraisal. *IEEE Access* 7, 114547–114571.
- West, D.B., 2001. *Introduction to graph theory*. Prentice hall, Upper Saddle River.
- Yuan, J., Hansen, I.A., 2007. Optimizing capacity utilization of stations by estimating knock-on train delays. *Transp. Res. Part B: Methodol.* 41 (2), 202–217.
- Zhang, D., Peng, Y., Zhang, Y., Wu, D., Wang, H., Zhang, H., 2021. Train Time Delay Prediction for High-Speed Train Dispatching Based on Spatio-Temporal Graph Convolutional Network. *IEEE Trans. Intell. Transp. Syst.* 23(3), 2434–2444.

- Zhang, D., Xu, Y., Peng, Y., Zhang, Y., Wu, D., Wang, H., Liu, J., Mohammed, S., Calvi, A., 2022. Prediction of Train Station Delay Based on Multiattention Graph Convolution Network. *Journal of Advanced Transportation*, 2022,1-12.
- Zhou, P., Chen, L., Dai, X., Li, B., Chai, T., 2020. Intelligent prediction of train delay changes and propagation using RVFLNs with improved transfer learning and ensemble learning. *IEEE Trans. Intell. Transp. Syst.* 22 (12), 7432–7444.
- Zilko, A.A., Kurowicka, D., Goverde, R.M., 2016. Modeling railway disruption lengths with Copula Bayesian Networks. *Transp. Res. Part C: Emerging Technol.* 68, 350–368.